The best frequency for inverse bremsstrahlung in excimer laser gases: a krypton fill, the measured 5s to 5p rate, and the pulse

Cooper K. Watson

October 12, 2026

1 The question and the answer

A wave of frequency f heats the electrons of an excimer laser gas by inverse bremsstrahlung. To deposit a power density P evenly through a layer of depth L, it has to be absorbed over that depth and not reflected at the surface. The best frequency f_{best} is the frequency that does both, defined in Section 2. It depends on the electrons through their temperature and density and on the atoms and ions they collide with, and the field that heats the electrons also excites the gas. The excited atoms are ionized by a second electron at under a third of the energy the ground state needs, they ionize each other and the krypton, and they trade energy with the electrons both ways. This note follows them inside both balances of the electrons, the balance of their number and the balance of their energy, at every point of a table of three fills, two buffered and one of krypton with 0.5 % F_2, five pressures and five electron densities, 75 points in all, and it does four things the earlier form of the calculation could not.

First, the excitation of krypton's 5s levels to 5p, the rate that carries most of the cooling of the electrons, is taken from the measured cross sections of Jung et al. [11, 12] rather than from a discharge model's fit, and the 5p manifold radiates to 5s at the measured lifetimes [29]. Second, the krypton fill is added, with no buffer gas. Its best frequency lies 1.04 to 1.20 times the argon fill's at the same pressure and density, 1.05 to 8.8 THz at 0.1 atm and 3.8 to 26.9 THz at 2 atm, with its electrons at 0.94 to 2.44 eV, and the beam holds the density at 11 of its 25 points. With the measured cross section the neon fill lies at 0.933 to 0.997 and the argon fill at 0.646 to 0.995 of the calculation without excited atoms. The fit in place of the measurement raises the best frequency by up to 5 %, the 5p rung's electron collisions left out raise it by up to 1.46 times, and the return of 5p to 5s by collisions, raised from the estimate used here to the one measured total loss rate of a 5p level [28], lowers it by up to 13 % at the dense krypton points and the electron temperature there by 19 %: that return rate is now the largest uncertainty of the answer. Third, the calculation is marched in time through a pulse of 1 μs at three points, with the F_2 burning and the density following its own balance. The steady table is the state of the first 100 ns at 10^{15} cm^{-3} and of the first 10 ns at 10^{16} cm^{-3}: the attachment that holds the density also burns the F_2, which is gone in 0.36 to 0.38 μs at 10^{15} cm^{-3} and in 35 ns at 10^{16} cm^{-3}, after which the density runs up 3.7 to 6.8 times to where the beam's pairs balance recombination, the electrons cool to 0.53 to 0.54 eV and the best frequency moves, by -18 and -6 % at 10^{15} cm^{-3} and by 2.8 times at 10^{16} cm^{-3}. Fourth, the kinetic solver is checked against a second solver of the same equation, written independently for a time-domain model of the KrF laser: at seven states from 10^{12} to 10^{16} cm^{-3} the two agree within 0.2 % in the electrons' mean energy and in the wave's absorbed power, and the two full models,

started from the same cold seed under the same wave, decay the plasma at rates 1 % apart with electrons 0.3 % apart in temperature.

2 The gas, the electrons and the best frequency

The table. The three fills are Ne/Kr/F_2 in the mole fractions 0.948/0.05/0.002, Ar/Kr/F_2 in 0.597/0.4/0.003 and Kr/F_2 in 0.995/0.005, at 300 K and pressures of 0.1, 0.2, 0.5, 1 and 2 atm. The electron density n_e is 10^{14}, 3×10^{14}, 10^{15}, 3×10^{15} or 10^{16} cm^{-3}, and the field deposits $P = 300$ kW/cm^3 in a layer of depth $L = 10$ cm. At each of these 75 points the frequency runs over 13 values, five per decade from 0.1 to 25.1 THz.

The electrons. The electrons' energy distribution $F(\epsilon)$ solves the two-term Boltzmann equation in the oscillating field [1], with the cross sections of Biagi's database for neon, argon, krypton and F_2 [2], electron–electron collisions, and collisions with the cations through the Coulomb cross section $\pi\,(e^2/4\pi\epsilon_0\epsilon)^2\Lambda(\epsilon,\omega)$. The logarithm Λ is the quantum one of Turnbull et al. for the photon energy $\hbar\omega$ [3, 4], cut off at the ion-sphere radius of the cations. The field's amplitude is set so that the electrons absorb P. The electrons' complex mobility $\mu(\omega)$ gives the conductivity $\sigma = n_e e\mu$ and the dielectric function $\varepsilon = 1 + i\sigma/\epsilon_0\omega$, and from them the absorption coefficient and the reflection at normal incidence,

$$\alpha = \frac{2\omega}{c}\,\mathrm{Im}\sqrt{\varepsilon}, \qquad R = \left|\frac{1-\sqrt{\varepsilon}}{1+\sqrt{\varepsilon}}\right|^2 . \tag{1}$$

α falls with f once ω exceeds the collision and plasma frequencies. f_A is the frequency at which $\alpha L = 1$, so that the wave is absorbed over the layer, and f_R the lowest at which $R \le 0.05$. The best frequency is

$$f_{\mathrm{best}} = \max\left(f_A,\, f_R\right), \tag{2}$$

both found by interpolating $\ln\alpha$ and $\ln R$ in $\ln f$ between the frequencies of the scan.

The ions and how the density is held. Electrons are lost to attachment on F_2 and to dissociative recombination on molecular cations. The two buffered fills have five cations each, the atomic and the molecular ion of the buffer gas, the heteronuclear ion of the buffer with krypton, Kr^+ and Kr_2^+, and the krypton fill has Kr^+ and Kr_2^+. They are made by the field's ionization of each gas and by an electron beam, they convert into one another, and they recombine with the electrons and with F^-, through the reactions of Table 10 in Appendix B. The recombination with F^- turns over with pressure, as the three-body capture calculations that Kannari et al. review do [7], and the other charges screen each pair, which multiplies it here by $S_c = 1/(1 + R_c/\lambda_D)$ with $R_c = 2e^2/(3\cdot4\pi\epsilon_0 kT_g)$ and λ_D the Debye length of all the charges at the gas temperature T_g. Where the ionization falls short of the losses, the beam makes up the difference and holds n_e. Its electrons enter with the energy spectrum of the slow secondaries of a fast beam, whose mean is 4.506 eV, and they are marked b where they bring more than a third of the absorbed power. Where the ionization exceeds the losses, the field multiplies the electrons at the net rate g, and the point is marked a where g exceeds one e-fold in the pulse of 1 μs. A point marked a is not steady over the pulse, and its best frequency is that of the density it passes through.

Table 1: The levels whose energy reaches the electrons, as formation energies above the ground-state atoms, in eV.

species	energy	origin
Kr^*	9.915	lowest 5s level, 9.9152, [6]
Kr^{**}	11.300	lowest 5p level, 11.3035, [6]
Kr^+	13.996	ionization limit, 13.9996, [6]
Ar^*	11.548	lowest 4s level, 11.5484, [6]
Ar^+	15.760	ionization limit, 15.7596, [6]
Ne^*	16.619	lowest 3s level, 16.6191, [6]
Ne^+	21.560	ionization limit, 21.5645, [6]
Kr_2^+	12.846	Kr^+ less the binding 1.15 measured, [19]
Ar_2^+	14.530	Ar^+ less the binding 1.23 measured, [19]
Ar_2^*	10.868	taken
Kr_2^*	9.200	taken
Ne_2^*	16.100	taken
Ne_2^+	20.200	taken
$ArKr^*$	9.500	taken
$ArKr^+$	13.500	taken
$NeKr^+$	13.700	taken

3 The excited atoms

The levels. The field excites krypton into its four 5s levels and into 5p and the levels above. They are followed as two species, Kr^* for the 5s levels and Kr^{**} for 5p and above, each standing at the energy of its lowest level, 9.915 and 11.30 eV, so that no excitation booked into a species creates energy. The buffer gas is followed the same way, Ar^* at argon's lowest 4s level, 11.548 eV, and Ne^* at neon's lowest 3s level, 16.619 eV, and four excited dimers form from these atoms, mostly in three-body collisions, Kr_2^*, Ar_2^*, $ArKr^*$ and Ne_2^*. The krypton fill carries Kr^*, Kr^{**} and Kr_2^*. Every energy is a formation energy above the ground-state atoms, Table 1. Those of the atoms lie within 0.005 eV of the levels and ionization limits of the NIST database [6], those of Kr_2^+ and Ar_2^+ follow from the measured binding of the two ions [19], and the others are taken as the table gives them.

The field makes Kr^* and Kr^{**} through the excitation cross sections of the Boltzmann equation, the 5s group into Kr^* and every higher group into Kr^{**}, and the buffer's excitations into its own species. The beam shares its energy between the gases by their stopping powers for fast electrons, from Bethe's formula at 600 keV with mean excitation energies of 137, 188, 352 and 115 eV for Ne, Ar, Kr and F_2, and makes one ion pair per 36.6 eV in neon [23], 26.2 eV in argon and 32 eV in F_2 [7], and 24.2 eV in krypton, where Kannari et al. print 24.3. With each pair it makes 0.305 excited atoms in neon and 0.284 in argon, the energies per pair of Lee et al. over their energies per excited atom, 120 and 93 eV [23], and 0.28 in krypton [7]. The recombination of the molecular cations leaves an excited atom behind. The excited atoms are lost to F_2, which turns Kr^* into KrF^* at 8.47×10^{-10} cm^3 s^{-1}, the mean of three measurements, with a yield of 0.90 [13, 14], to the three-body formation of dimers, to radiation, to Penning ionization, and to the electrons. Tables 9 and 10 of Appendix B list every reaction between heavy particles with its rate and its source, and Table 2 the electron collisions. Each species stands at the steady state of its balance, solved for all of them at once at the electrons' distribution. Where the field multiplies the electrons at the rate g, every density grows with them, the ions as the excited atoms, and each species is diluted at gn on top of its losses.

Table 2: The electron collisions of the excited atoms as the Boltzmann equation carries them. The rate is the source's in $\mathrm{cm}^3\,\mathrm{s}^{-1}$ with T_e in eV, E_0 the onset of the cross section of Eq. (3) in eV, and Δ the energy the electron loses, negative where it gains. The standing says whether the rate is measured, printed by its source from elsewhere, calculated, the value of a related process, estimated, or detailed balance.

reaction	rate	E_0	Δ	standing	source
$e + \mathrm{Kr}^* \rightarrow \mathrm{Kr}^+ + 2\,e$	$7.00 \times 10^{-8}\,T_e^{0.68} e^{-4.04/T_e}$	4.081	+4.081	calculated	the first fits the calculation of the second, [7, 9]
$e + \mathrm{Kr}^* \rightarrow e + \mathrm{Kr}^{**}$	measured cross section, Sec. 3	1.385	+1.385	measured	the two metastable 5s levels into the ten 5p levels, lumped over the 5s manifold 5:3:1:3 (Sec. 3), [11, 12]
$e + \mathrm{Kr}^{**} \rightarrow e + \mathrm{Kr}^*$	Eq. (4), $g_l/g_u = 1/3$	0.000	-1.385	detailed balance	of the measured excitation, [5]
$e + \mathrm{Kr}^{**} \rightarrow \mathrm{Kr}^+ + 2\,e$	$2.00 \times 10^{-7}\,T_e^{0.62} e^{-2.53/T_e}$	2.696	+2.696	calculated	[7, 9]
$e + \mathrm{Kr}^* \rightarrow e + \mathrm{Kr}$	Eq. (4), $g_l/g_u = 1/12$	0.000	-9.915	detailed balance	of the 5s excitation, [5, 2]
$e + \mathrm{Ar}^* \rightarrow \mathrm{Ar}^+ + 2\,e$	$1.56{\times}10^{-7}\,T_e^{0.71} e^{-4.212/T_e}$	4.212	+4.212	analog	the law R9 of the source for its higher excited argon, at the 4s binding energy, [25]
$e + \mathrm{Ar}^* \rightarrow e + \mathrm{Ar}$	Eq. (4), $g_l/g_u = 1/12$	0.000	-11.548	detailed balance	[5, 2]
$e + \mathrm{Ar}^* \rightarrow e + \mathrm{Ar}$	Eq. (4), $g_l/g_u = 1/12$	0.000	-11.623	detailed balance	[5, 2]
$e + \mathrm{Ar}^* \rightarrow e + \mathrm{Ar}$	Eq. (4), $g_l/g_u = 1/12$	0.000	-11.723	detailed balance	[5, 2]
$e + \mathrm{Ar}^* \rightarrow e + \mathrm{Ar}$	Eq. (4), $g_l/g_u = 1/12$	0.000	-11.828	detailed balance	[5, 2]
$\mathrm{ArKr}^* + e \rightarrow \mathrm{ArKr}^+ + 2\,e$	$9.00 \times 10^{-8}\,T_e^{0.7} e^{-3.6/T_e}$	4.000	+4.000	printed	[7]
$\mathrm{ArKr}^* + e \rightarrow \mathrm{Ar} + \mathrm{Kr} + e$	1.0×10^{-7}	0	-9.500	estimated	estimated there, [7]
$e + \mathrm{Ne}^* \rightarrow \mathrm{Ne}^+ + 2\,e$	$3.04 \times 10^{-8}\,T_e^{0.54} e^{-7.5/T_e}$	7.500	+4.941	printed	[23]
$e + \mathrm{Ne}_2^* \rightarrow \mathrm{Ne}_2^+ + 2\,e$	$9.75 \times 10^{-9}\,T_e^{0.71} e^{-3.4/T_e}$	4.100	+4.100	printed	[25]

The 5s to 5p excitation is measured. Jung et al. measured the cross sections for electron-impact excitation out of krypton's two metastable 5s levels, $1s_5$ and $1s_3$ in Paschen's notation, into each of the ten 5p levels, from 2 to 300 eV, with uncertainties of 35 % for $1s_5$ and 40 % for $1s_3$ [11, 12]. The cross section used here for $\mathrm{Kr}^* \to \mathrm{Kr}^{**}$ is their sum over the ten 5p levels, lumped over the 5s manifold by statistical weight, $1s_5 : 1s_4 : 1s_3 : 1s_2 = 5 : 3 : 1 : 3$, with each unmeasured resonance level taken as its core partner, $1s_4$ as $1s_5$ and $1s_2$ as $1s_3$, since the core-preserving cross sections scale with the optical oscillator strengths. Each level's cross section is zero below its threshold, rises linearly to the first tabulated point, follows the table log-log between the points, and falls as $1/\epsilon$ past the last. Its onset on the ladder is the gap between the lumped levels, 1.385 eV. On Maxwellians this cross section gives 1.36 times the rate of the discharge model's fit that Kannari et al. print [7, 10] at the coldest electrons of the table, 0.87 eV, and 0.92 times at the hottest, 4.69 eV, differences at or inside the measurement's own uncertainty. The fit is kept as the upper end of a band, Section 6.

The 5p manifold returns to 5s three ways. Kr^{**} gives its energy back to the electrons in superelastic collisions, the detailed balance of the excitation, and it returns to 5s by radiation and in collisions with the gas. The ten 5p levels radiate to the 5s levels in the near infrared, on lines that end on the 5s levels and are not trapped, with the lifetimes that Dzierzega et al. derive from the beam-gas-laser measurements of six of the levels, 23.1 to 38.1 ns [29]. The lumped manifold decays at the $2J + 1$-weighted mean of their rates, one lifetime of 29.0 ns, and the 1.385 eV leaves as light. The return by collisions is the estimate of 1.0×10^{-10} $\mathrm{cm^3\,s^{-1}}$ that Kannari et al. mark as such for Kr^{**} + Ar [7], used for every partner. One measurement in krypton exists: Gazeli et al. measured the total loss rate of the $5p'[3/2]_2$ level in krypton at 300 K, $(1.7 \pm 0.2) \times 10^{-10}$ $\mathrm{cm^3\,s^{-1}}$, without resolving its product [28]. It bounds the return of the lumped manifold from above and is run as the upper end of a band. The level-resolved rates of Whitehead et al. [30] would settle it and are not used here. At the krypton fill at 1 atm and 10^{15} $\mathrm{cm^{-3}}$, at 10 THz, the manifold returns 84 % by collisions, 15 % superelastically and 1.2 % by radiation.

Stepwise ionization has no measurement. No measurement of the ionization of krypton's 5s levels by electrons was found, as a cross section or as a rate. Kannari et al. print $7.00 \times 10^{-8}\, T_e^{0.68} e^{-4.04/T_e}$ $\mathrm{cm^3\,s^{-1}}$ for it from Vriens's binary-encounter calculation [7, 9], and Johnson and Hunter print 4.8×10^{-8} from the same calculation [8]. This note takes Kannari's fit. The stepwise ionization of Ar^* has only a transplanted law, that of Chen et al. for the higher of their two excited argon species [25] at argon's 4s binding energy. In the calculation with Kannari's 5s to 5p fit, a stepwise rate 2.1 to 2.3 times Kannari's, or none at all, moved the best frequency by less than 1.5 %, so the answer does not rest on this rate and no band is run on it here.

4 The excited atoms inside the electrons' balances

Every electron collision of an excited atom is a process of the Boltzmann equation. Each excited species enters the Boltzmann equation as a gas of its own, at the mole fraction its balance gives, with every collision of an electron with it that Table 2 lists. A rate fitted as $k = A\,T^B e^{-E/T}$ becomes the cross section

$$\sigma(\epsilon) = C\,\frac{(\epsilon - E_0)^{B+1/2}}{\epsilon} \quad (\epsilon > E_0), \qquad C = \frac{A\sqrt{\pi}}{2\gamma\,\Gamma(B + 3/2)}, \tag{3}$$

with $\gamma = (2e/m_e)^{1/2}$, whose rate over a Maxwellian of temperature T is exactly $AT^B e^{-E_0/T}$. The electron loses in it the energy Δ of the reaction on the ladder of formation energies of Table 1. An electron below Δ cannot pay it, so the onset E_0 is the larger of the fit's E and Δ. Where the fit's E lies above Δ the printed rate is kept and the electron loses Δ. Where it lies below, the onset moves up to Δ, which lowers the rate on a Maxwellian by $e^{-(\Delta-E)/T}$. A constant rate k is the cross section $k/\gamma\sqrt{\epsilon}$, and a measured cross section enters as it is, above the ladder's gap.

De-excitation by detailed balance. In a superelastic collision the electron gains the energy Δ of the level it empties. Its cross section follows from that of the excitation it reverses by detailed balance [5],

$$\sigma_\downarrow(\epsilon) = \frac{g_l}{g_u}\,\frac{\epsilon+\Delta}{\epsilon}\,\sigma_\uparrow(\epsilon+\Delta), \tag{4}$$

with g_l and g_u the statistical weights of the lower and the upper level. Over a Maxwellian its rate is $(g_l/g_u)\,e^{\Delta/T}$ times that of the excitation, and when the electrons are in equilibrium with the excited atoms the two flows cancel. The Boltzmann operator places the electron Δ above the energy it had, so that the equation is no longer solved from the top of the energy grid down and is solved as one linear system.

The electrons that the excited atoms set free. Penning ionization, between two excited atoms or dimers or between one of them and ground-state krypton, releases an electron with the energy that the reaction leaves on the ladder, from 2.1 to 12.0 eV. These electrons enter the Boltzmann equation as a source at their release energy, each in a band 0.2 eV wide.

The two closures. The Boltzmann equation is solved in steady state with a source of electrons. Where the beam holds the density, the source is the beam's slow secondaries and the Penning electrons, mixed in proportion to their rates, and the rate of injection per electron is what the losses leave over,

$$s = N\,(k_{\mathrm{att}} + k_{\mathrm{rec}} - k_{\mathrm{ion}}) = \frac{S_{\mathrm{beam}} + S_{\mathrm{pen}}}{n_e}, \tag{5}$$

with N the gas density and k_{ion} the ionization of every gas of the equation, the excited ones among them. Where the field multiplies the electrons at the rate g, the Penning electrons are the source and g enters as a loss from every energy in proportion to the electrons there, the dilution that growth brings, set so that

$$N\,(k_{\mathrm{att}} + k_{\mathrm{rec}} - k_{\mathrm{ion}}) + g = \frac{S_{\mathrm{pen}}}{n_e}. \tag{6}$$

Without a source the second closure is the growth model of the two-term equation [1]. It leaves out the share of the growth term in the anisotropic part of the distribution, which is of the order of g/ν_m, the growth rate over the momentum-transfer frequency, below 3×10^{-3} at every growing point. The cations follow the same particle balance, with the excited atoms' stepwise electrons added to the ionization of their ground gas and the dimers' and the Penning electrons to their own cations.

The loop. Each round solves the Boltzmann equation with the excited atoms at their current mole fractions, then their balance with the rates over the new distribution, then the ions, and sets the closure and the source of the next round. The mole fractions are relaxed in their logarithm, and the loop ends when they, the ions and the beam's or the growth rate change by less than 2×10^{-3} between two rounds and the closure no longer changes. A growth rate's change is measured against

Table 3: The best frequency f_{best} in THz at $L = 10$ cm and 300 kW/cm^3 with the excited atoms inside the electrons' balances. Marked a where the field multiplies the electrons more than e-fold in the pulse of 1 μs, and b where the beam's electrons bring more than a third of the absorbed power.

fill	p [atm]	$n_e = 10^{14}$	3×10^{14}	10^{15}	3×10^{15}	10^{16} cm^{-3}
Ne	0.1	0.64^a	1^a	1.7^a	2.9^a	5.7^a
Ne	0.2	0.86^a	1.4^a	2.3^a	3.8^a	7.2
Ne	0.5	1.3^a	2.1^a	3.4^a	5.7	10^b
Ne	1	1.7^a	2.8^a	4.7	7.7^b	14^b
Ne	2	2.4^a	3.9	6.6	11^b	19^b
Ar	0.1	1^a	1.7^a	2.9^a	4.6^a	8.2
Ar	0.2	1.3^a	2.2^a	3.8^a	6	10^b
Ar	0.5	1.9^a	3	5.3	8.6^b	15^b
Ar	1	2.6	4.4	7.5^b	11^b	17^b
Ar	2	3.5	6.3	11^b	16^b	23^b
Kr	0.1	1^a	1.8^a	3.1^a	5^a	8.8^b
Kr	0.2	1.4^a	2.3^a	4	6.6	11^b
Kr	0.5	–	3.4	6	9.4^b	16^b
Kr	1	2.7	4.9	8.6^b	13^b	19^b
Kr	2	3.8	7.1^b	13^b	19^b	27^b

the larger of itself and 10^6 s^{-1}, one e-fold in the pulse, below which a growth rate multiplies nothing and its relative change says nothing. At the boundary between the closures the two can alternate from round to round, with the beam's rate alternating between zero and a finite value and the excited atoms differing by up to 1 % between the two states, which is one and the same plasma within that: a point whose excited atoms and ions have settled within ten tolerances and whose closure has alternated for three rounds is accepted and flagged as a boundary point. It took 3 to 72 rounds per point, the most at the one boundary cell of Section 6. Each frequency starts from the state of the one below it. The electrons and the excited atoms book the same energy for every collision, which Appendix A tests row by row as an identity.

5 Results

The table. Table 3 and Figure 1 give the best frequency at the 75 points with the excited atoms inside the electrons' balances and the measured 5s to 5p cross section, and Table 11 of Appendix C every point with its electron temperature, its marks, the mole fraction of Kr*, the net energy the excited atoms take from the electrons, and the bands of Section 6. One cell of the krypton fill, 0.5 atm and 10^{14} cm^{-3}, is a marginal plasma whose two closures alternate at most of its frequencies, with the excited atoms moving by more than the boundary rule allows: 9 of its 13 frequencies closed, 4 of them at the flipping boundary, and the cell is reported incomplete, blank in the table and the figure and left out of the bands.

The krypton fill against the argon fill. At the same pressure and density the krypton fill's best frequency lies 1.04 to 1.20 times the argon fill's, the least at 0.2 atm and 3×10^{14} cm^{-3} and the most at 2 atm and 3×10^{15} cm^{-3}. The reason is the collision frequency. Above the collision

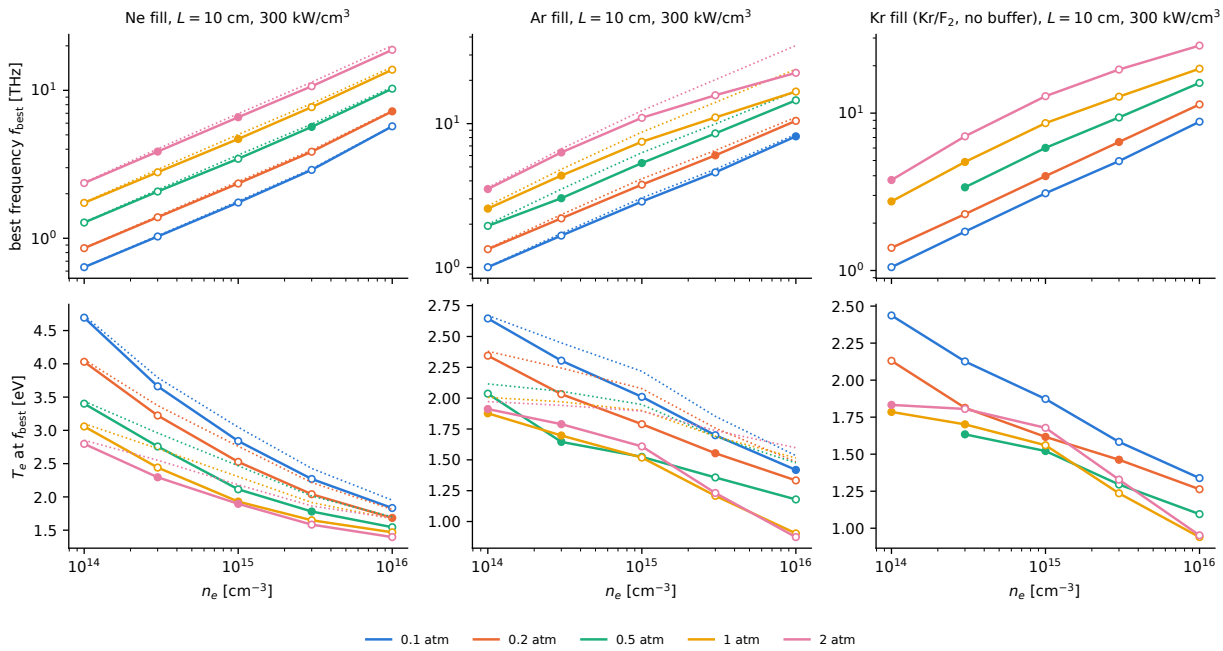


Figure 1: The best frequency at $L = 10$ cm and 300 kW/cm^3 against n_e, top, and the electron temperature $2\langle\epsilon\rangle/3$ there, bottom, with the excited atoms as solid lines and, for the two buffered fills, without them dotted. An open marker is a point marked a or b in Table 3. The krypton fill's cell at 0.5 atm and 10^{14} cm^{-3} is left blank.

frequency α falls as ν_m/ω^2, so the frequency at which $\alpha L = 1$ scales as the square root of the effective momentum-transfer frequency $\nu_m = \omega\mu_r/\mu_i$, the ratio of the real to the imaginary part of the mobility times ω. At the best frequencies ν_m of the krypton fill is 1.08 to 1.52 times the argon fill's, and its square root gives the ratio of the best frequencies within 3.5 %. Krypton's momentum-transfer cross section rises from 0.54 of argon's at 1 eV to 1.55 times at 3 eV [2], and the argon fill is 40 % krypton. The electrons of the krypton fill sit at 0.94 to 2.44 eV, those of the argon fill at 0.87 to 2.65 eV and those of the neon fill at 1.40 to 4.69 eV.

How the density is held. The field multiplies the electrons at 15 points of the neon fill, 8 of the argon fill and 6 of the krypton fill, marked a, and the beam brings more than a third of the absorbed power at 5, 9 and 11, marked b. The excited atoms take from the electrons, net of what they give back in superelastic collisions, 0.007 to 2.9 times the absorbed power in the neon fill, 0.012 to 15.0 times in the argon fill and 0.017 to 23.5 times in the krypton fill, the most at 2 atm and 10^{16} cm^{-3} in each. Where they take more than the field gives, the beam holds the density and its slow electrons bring the difference. With the measured cross section the best frequency of the neon fill lies at 0.933 to 0.997 of its value without the excited atoms and that of the argon fill at 0.646 to 0.995, falling at every point, by the most at 2 atm and 10^{16} cm^{-3} in the argon fill, from 34.9 to 22.5 THz. The electrons there are at 0.55 to 0.99 of their temperature without the excited atoms in the argon fill and 0.83 to 0.99 in the neon fill. The krypton fill has no counterpart without excited atoms.

6 What limits the answer

Three bands. Three more scans of the 75 points bound what the rates of the 5p rung can do, Table 11. With Kannari's fit in place of the measured 5s to 5p cross section the best frequency rises to 1.000 to 1.049 times the reference, up to 1.010 in the neon fill, 1.047 in the argon fill and 1.049 in the krypton fill, the most at 2 atm and 10^{16} cm^{-3}, where the krypton fill moves from 26.9 to 28.2 THz, with the electrons hotter by at most 6.7 %. With every electron collision of the 5p rung left out, so that collisions and radiation alone return Kr** to 5s, it rises to 1.00 to 1.46 times the reference, up to 1.054 in the neon fill, 1.460 in the argon fill and 1.436 in the krypton fill, the most at 2 atm and 10^{16} cm^{-3} in the argon fill, where it returns to 32.9 THz: the 5s to 5p cycle carries the fall. With the return of 5p to 5s by collisions raised from Kannari's estimate to the measured total loss rate of Gazeli et al., 1.7×10^{-10} cm^3 s^{-1}, it falls to 0.868 to 1.000 times the reference, to 0.999 in the neon fill, 0.951 in the argon fill and 0.868 in the krypton fill, the most at 0.5 atm and 10^{16} cm^{-3} in the krypton fill, where it moves from 15.6 to 13.5 THz with the electrons at 0.81 of their reference temperature. A faster return empties the 5p manifold before the electrons can ionize it or take their energy back, so more of the energy of the cycle ends in the gas and the electrons are colder. The return rate by collisions is thus the largest uncertainty of the answer at the dense krypton points, 13 % in the best frequency and 19 % in the electron temperature, and a level-resolved measurement in krypton [30] would replace it.

What the model leaves out. Kr** lumps 5p and the levels above at the lowest 5p level, and argon and neon have no second rung. The four argon levels give back up to 0.28 eV more in superelastic collisions than the lumped Ar* holds. The growing closure leaves out a share of order $g/\nu_m < 3 \times 10^{-3}$ in the anisotropic part of the distribution, one point of the argon fill lies on the boundary between the closures and one cell of the krypton fill is incomplete. The recombination of Kr_2^+ puts the sum of the two branches that Kannari et al. print at 300 K on a $T_e^{-1/2}$ law [7]. The excited atoms settle within 220 ns, but the gas is held at its fill composition, while attachment alone would take the F_2 down e-fold in 24 to 56 ns at 10^{16} cm^{-3} and in 0.44 to 0.97 μs at 10^{15} cm^{-3}. What that means for the table is the subject of the next section.

7 The pulse

The march. The steady state of the table is marched in time at three of its points. The kinetic solution of the electrons is renewed every few nanoseconds: whenever the slowest of the electron density, the F_2 and the excited atoms would move by 2 %, at least every 10 ns, 5 ns in the third pulse, until a twentieth of the time elapsed exceeds that, and every twentieth of the time elapsed after, and between renewals the excited species, the cations, F^- and F_2 are integrated with the renewed rate coefficients held, the electron density following from quasineutrality. The F_2 burns by attachment, by the harpoon reactions of the excited atoms and by its own ionization. At each renewal the field is solved again so that the wave deposits P, the solve carries the electrons' sources, the beam's secondaries and the Penning electrons at their release energies, and the instantaneous net rate g of the density enters as the uniform loss of Eq. (6), given rather than solved, so that the distribution's shape is that of a plasma changing at that rate. The beam is a constant current, at the pair rate that held the density at the point's steady state at $t = 0$. Held still are the gas temperature, 300 K, the atomic fluorine beyond the consumption of F_2, the excimers, which are sinks, and the buffer and krypton densities. At chosen times the 13-frequency scan is solved at the frozen state and the best frequency of that state found by the definitions of Section 2.

Table 4: Three pulses at 300 kW/cm^3, the beam holding the density at $t = 0$: the electron density in cm^{-3}, the electron temperature in eV, the mole fraction of Kr* in units of 10^{-4}, the F_2 over its fill value, the net rate g of the density in 10^6 s^{-1}, the rms field in kV/cm and the absorption coefficient in m^{-1} at the wave's frequency.

t [ns]	n_e	T_e	x_{Kr^*}	F_2	g	E	α
Kr fill, 1 atm, 10^{15} *cm*$^{-3}$*, 10 THz*							
0	1.00×10^{15}	1.560	0.74	1.000	0.00	38.9	7.48
51	1.11×10^{15}	1.476	0.85	0.798	1.65	38.3	7.71
100	1.25×10^{15}	1.370	1.02	0.613	1.50	38.2	7.76
198	1.61×10^{15}	1.081	1.71	0.288	1.56	40.5	6.91
300	2.36×10^{15}	0.728	3.85	0.060	4.16	46.9	5.14
489	3.74×10^{15}	0.541	7.62	0.000	0.21	47.3	5.05
1000	3.75×10^{15}	0.535	7.70	0.000	0.00	47.6	4.99
Ar fill, 1 atm, 10^{15} *cm*$^{-3}$*, 7.7 THz*							
0	1.00×10^{15}	1.519	0.72	1.000	0.00	34.6	9.44
49	1.12×10^{15}	1.442	0.80	0.797	2.11	33.9	9.82
100	1.28×10^{15}	1.336	0.94	0.601	2.34	33.6	10.03
198	1.68×10^{15}	1.074	1.44	0.281	2.19	34.5	9.50
300	2.41×10^{15}	0.763	2.80	0.066	3.95	37.6	8.00
489	3.88×10^{15}	0.554	5.60	0.000	0.54	36.8	8.35
1000	3.95×10^{15}	0.542	5.81	0.000	0.00	37.0	8.29
Kr fill, 1 atm, 10^{16} *cm*$^{-3}$*, 19.2 THz*							
0	1.00×10^{16}	0.943	5.09	1.000	0.00	33.6	10.04
10	1.32×10^{16}	0.838	6.42	0.536	29.93	31.8	11.20
20	1.97×10^{16}	0.691	10.73	0.193	41.35	28.7	13.77
30	3.22×10^{16}	0.599	17.89	0.028	48.70	21.9	23.64
49	5.58×10^{16}	0.554	28.97	0.000	14.87	14.4	55.17
100	6.86×10^{16}	0.537	33.90	0.000	0.35	12.0	78.62
300	6.84×10^{16}	0.533	34.12	0.000	-0.01	12.0	78.54
1000	6.83×10^{16}	0.532	34.13	0.000	0.00	12.1	78.38

Table 5: The best frequency in THz of the frozen state at the checkpoints of the three pulses, with f_R in parentheses.

pulse	0 ns	30 ns	100 ns	300 ns	1000 ns
Kr fill, 1 atm, 10^{15} cm^{-3}, 10 THz	8.64 (0.23)	–	8.80 (0.31)	7.18 (0.55)	7.09 (0.72)
Ar fill, 1 atm, 10^{15} cm^{-3}, 7.7 THz	7.48 (0.27)	–	7.71 (0.34)	6.89 (0.56)	7.03 (0.75)
Kr fill, 1 atm, 10^{16} cm^{-3}, 19.2 THz	19.19 (1.16)	29.11 (2.18)	52.86 (3.17)	52.83 (3.17)	52.77 (3.16)

What the pulses say. Table 4 follows the krypton fill and the argon fill at 1 atm and 10^{15} cm^{-3} at the table's best frequencies, 10 and 7.7 THz, and the krypton fill at 1 atm and 10^{16} cm^{-3} at its own, 19.2 THz, and Table 5 the best frequency of the frozen state at the checkpoints. The attachment that holds the density burns the F_2 at 3×10^8 to 6×10^8 molecules per electron per second. At 10^{15} cm^{-3} half of it is gone at 134 ns and all but 1 % by 365 ns in the krypton fill and 383 ns in the argon fill. The density then runs up, 3.75 times in the krypton fill and 3.95 times in the argon fill, until the beam's pairs balance recombination, which they do by 0.44 to 0.47 μs, the same power spread over four times the electrons cools them from 1.56 and 1.52 eV to 0.535 and 0.542 eV, Kr* accumulates 10 and 8 times once its harpoon partner is gone, and the field for the same deposited power rises from 38.9 to 47.6 kV/cm in the krypton fill. The best frequency of the plasma rises at first, by 2 and 3 % at 100 ns, then falls, to 0.82 and 0.94 of its start at 1 μs. At 100 ns the density is up by 25 and 28 %, the electrons are at 1.37 and 1.34 eV and the F_2 at 0.61 and 0.60 of its fill, so the state of the table holds, within 3 % in the best frequency, for the first 100 ns at this density. At 10^{16} cm^{-3} half of the F_2 is gone at 11 ns and all but 1 % at 35 ns, the density runs up 6.8 times by 92 ns, the electrons cool from 0.94 to 0.53 eV, the field for the same power falls from 33.6 to 12.1 kV/cm as α at 19 THz climbs from 10.0 to 78 m^{-1}, so that the wave then stops within 1.3 cm of the 10-cm layer, and the best frequency of the new plasma is 1.52 times the table's at 30 ns and 2.75 times, 53 THz, from 100 ns on. At 10 ns the density is up by 32 % and the F_2 at 0.54 of its fill: at 10^{16} cm^{-3} the table's answer holds for the first 10 ns. A shorter pulse, a lower beam current or more F_2 keeps the table's answer longer, and the march gives the time it holds at any point.

8 The kinetic solver against a second solver

The same equation twice. The two-term Boltzmann equation of Section 2, with its wave term, its electron–electron term, its growth closure and the electron–ion absorption of Turnbull et al., was solved a second time by a solver written independently for a time-domain model of the KrF laser, with its own implementation of the Turnbull construction from the same two papers [3, 4] and the same cross sections [2]. Table 6 compares the two on the krypton fill at seven states, from 10^{12} to 10^{16} cm^{-3}, 0.1 to 1 atm and 95 GHz to 10 THz, with the ions carrying from none to a quarter of the absorption: a given rms field and frequency, no excited atoms, no sources, the net frequency of ionization and attachment self-consistent in both, the second solver started from the first's converged distribution. The mean energies agree within 0.2 %, the wave's absorbed power within 0.2 % and the net frequency within 0.6 % at every state. The share of the absorption the two book to the ions differs by definition at 95 GHz, 0.19 against 0.17 at 0.1 atm and 10^{16} cm^{-3}, while the total power agrees to 0.02 %: the first books the ions' part by their share of the momentum transfer in the field term, the second re-evaluates the field term with the atoms' momentum transfer alone, and below $\omega \sim \nu_m$ that is not the complement. The optics of Eq. (1) computed on the second solver's distributions, from the same momentum-transfer sum its wave term uses, give α within 0.2 % and R within 0.4 % of the first's at the seven states, over α from 2.3 to 3.4×10^4 m^{-1} and R from 3×10^{-8} to 0.91, and the power the mobility implies equals the second solver's own absorbed power to 10^{-4}. The absorption that defines the best frequency is therefore the same in both.

The two full models from the same cold start. The second solver belongs to a time-domain model of the KrF laser that evolves the gas temperature, the excimers, the photons and the walls as well as the species followed here. The two full models were started from one and the same state, the krypton fill at 1 atm with a seed of 10^{15} electrons/cm^3 and as many Kr$^+$, no excited atoms, no

Table 6: The two kinetic solvers at one and the same state of the krypton fill: the rms field in kV/cm, the reduced field in Td, the mean energy in eV, first solver / second solver, the ratio of the wave's absorbed power per electron, second over first, the share of the absorption each books to the ions, and the ratio of the net frequency of ionization and attachment, second over first.

p [atm]	n_e [cm^{-3}]	E	f	E/N	$\langle\epsilon\rangle$	P_2/P_1	ions' share	ν_2/ν_1
1	10^{12}	1.5	95 GHz	6.1	3.8061 / 3.8083	0.9994	0.0000 / 0.0000	0.9999
1	10^{14}	1.5	95 GHz	6.1	3.5558 / 3.5580	0.9990	0.0009 / 0.0005	1.0032
1	10^{13}	3	1 THz	12.3	3.1950 / 3.1995	1.0014	0.0000 / 0.0000	0.9984
1	10^{15}	3	1 THz	12.3	2.5933 / 2.5964	1.0011	0.0024 / 0.0021	1.0045
1	10^{15}	30	10 THz	122.6	2.5825 / 2.5854	1.0016	0.0013 / 0.0013	1.0047
0.1	10^{16}	2	1 THz	81.8	1.8584 / 1.8614	1.0016	0.2496 / 0.2483	1.0046
0.1	10^{16}	0.6	95 GHz	24.5	2.3229 / 2.3242	1.0002	0.1928 / 0.1719	1.0058

Table 7: The two full models from the same cold start, the krypton fill at 1 atm under 38.9 kV/cm at 10 THz with no beam: the electron density in cm^{-3}, the electron temperature in eV and the Kr* density in cm^{-3}, each second model / march, the F_2 over its fill value in both, and the second model's gas temperature in K.

t [ns]	n_e, second	march	T_e, second	march	Kr*, second	march	F_2, second	march	T_g
0	1.00×10^{15}	1.00×10^{15}	0.026	1.795	0	2.4×10^{7}	1.0000	1.0000	300.0
10	5.18×10^{11}	7.55×10^{13}	1.744	1.661	7.9×10^{12}	3.7×10^{14}	0.9915	0.9899	302.6
20	5.69×10^{10}	6.03×10^{12}	1.747	1.696	2.8×10^{12}	1.5×10^{14}	0.9914	0.9871	302.8
50	8.44×10^{7}	7.13×10^{9}	1.746	1.740	9.8×10^{10}	5.1×10^{12}	0.9914	0.9858	303.2
100	1.72×10^{3}	1.73×10^{5}	1.747	1.742	3.5×10^{8}	2.0×10^{10}	0.9914	0.9856	303.4

F^-, under a wave of 38.9 kV/cm rms at 10 THz, the field the table needs for 300 kW/cm^3 at this point, and no beam, and run for 100 ns, Table 7. Without a beam, attachment outruns the wave's own ionization at this fill and both models decay the plasma: the second model at 2.17×10^8 s^{-1} between 10 and 100 ns, the march of Section 7 at 2.15×10^8 s^{-1} from 40 ns on, 1 % apart, with the electrons of the decaying plasma at 1.747 and 1.742 eV, 0.3 % apart. The two differ only in the first 10 ns, where the second model heats its seed from the gas temperature and loses three decades of density to the attachment of the still-cold electrons on the way, while the march starts from the distribution solved hot at $t = 0$. The second model burns 0.9 % of the F_2, the march 1.4 %, the difference again in the first 10 ns, and the gas warms by 3.4 K in the second model while the march holds it at 300 K.

9 Where the calculation stands

The best frequency of Table 3 rests on the measured 5s to 5p cross section, on the measured lifetimes of the 5p levels, on an estimate for the return of 5p to 5s by collisions that one measured total loss rate bounds, and on a calculated stepwise ionization that the answer does not rest on. Its largest uncertainty is the return rate, 13 % in the best frequency and 19 % in the electron temperature at the dense krypton points, and a level-resolved measurement would remove it. The table is the state of the first 100 ns of a beam-held pulse at 10^{15} cm^{-3} and of the first 10 ns at 10^{16} cm^{-3}, after which the F_2 is gone and the best frequency of the plasma that remains is the march's, not the table's. The kinetic solver behind both agrees with an independent one to 0.2 %. What the calculation still

leaves out, the gas temperature, the excimers' own kinetics and the photons, the atomic fluorine and the walls, belongs to the time-domain model the second solver was written for, and the next step is to carry the absorption of Eq. (1) and the excited atoms' electron collisions of Table 2 into it.

A The checks

Every number of this note and every step of the calculation behind it is tested by the checks listed in Table 8, which end with all of them passing. Section K tests the electron collisions of the excited atoms: each fitted process against its source's printed rate on Maxwellians, the measured cross section against its tables and its own integral, each de-excitation against detailed balance, every energy against the ladder, the 5p lifetime against the ten level lifetimes, and the identity that the energy the excited atoms take from the electrons equals the sum, over the collisions the Boltzmann equation carries, of their events times the energy each takes. Section C checks every point of the four scans: each must close the same way for the electrons and the ions, the two energy books must agree, the residuals must be small, and one point is solved again in a fresh process. Section B recomputes a best frequency from the scan. Section L checks the comparison with the second solver and section P the pulses, their quasineutrality, the monotone burn of the F_2 and their checkpoints. Section N computes every number the text prints from the data, section E ties each printed number and claim to them, and section S checks the style and that no file or unpublished note is named.

Table 8: The checks of this note.

	check	what	value	
1	`K.rows_added`	the added rows are on: the 5p manifold's radiative decay among the rows, the return band	V4 True, PRAD rows [('DZIERZEGA', 'PRAD')], KPS_Kr band {'default': 1e-16, 'lo': 3e-17, 'hi': 1.7e-16, 'unit': 'm3/s', 'marks': ((1.7e-16, "Gazeli 2021, the ...	pass
2	`K.fit_maxwell`	each fitted process on Maxwellians (T 1, 2, 4 eV) = the source's printed rate x exp(-(onset - E)/T), onset max(E, gap), the electron losing the ledger's gap (three fills, Kannari's stepwise fit)	10 processes, largest 2.5e-05 at ('Ar', 'KSTEP', 1.0), wrong energies []	pass
3	`K.jung_tables`	the measured 5s → 5p cross sections: the tables read back at the tabulated energies, the lowest threshold against the ledger's gap, the fit's pseudo cross section returning the fit	Table II read back at the tabulated energies: [np. float64(27.0), np. float64(47.0), np. float64(52.0), np. float64(52.0), np. float64(51.0), np. float64(49.0)]; T...	pass
4	`K.jung_lumped`	the solver's K76 at 3 eV = the tables' sums weighted 8:4 by hand (1e-16 cm^2), a measured cross section above the ledger's gap	99.19 vs 99.19, gap 1.385	pass

	check	what	value	
5	K.jung_rate	the solver's rate of the measured K76 on Maxwellians (1, 2, 4 eV) = the module's own integral	largest relative difference 2.4e-05	pass
6	K.jung_reverse	its reverse by detailed balance at 2 eV: k_down/k_up = g_5s/g_5p exp(gap/T)	0.66624 vs 0.66624	pass
7	K.down_db	every de-excitation (Kr* and the four Ar levels to the ground, Kr** to Kr*) is its excitation's detailed balance on Maxwellians, at the energy the excitation takes	30 values, largest relative difference 1.8e-06	pass
8	K.const	ArKr* + e → Ar + Kr + e as sigma = k/(gamma sqrt(eps)) gives the printed constant on Maxwellians	k 1.000e-13 m^3/s, largest relative difference 4.4e-16	pass
9	K.ledger	every process the solver carries takes the ledger's energy of its row (the Ar superelastic rows give each level's own, above the lumped Ar*), and every lumped excited atom stands at or below the lowest field excitation booked into it	off the ledger [], Ar levels above Ar* by [-0.0, 0.075, 0.175, 0.28] eV, level vs lowest excitation {'Ar*': (11.548, 11.548), 'Kr*': (9.915, 9.915), 'Kr**': ...	pass
10	K.prad	Kr** → Kr* + light: a one-body row at 1/29.0 ns, the 2J+1-weighted mean decay rate of the ten semi-empirical 5p lifetimes of Dzierzega et al. (their Table 2, 23.1 to 38.1 ns; the row's energy on the ledger leaves as light)	row 3.4483e+07 /s = 1/29.00 ns, the ten give 29.04 ns	pass
11	K.kps_band	the 5p → 5s return in collisions with Kr: Kannari's estimate 1.0e-10 cm^3/s as the default, Gazeli et al.'s measured total loss 1.7e-10 as the band's top, and the four scans' settings (the reference with the measured K76, the fit, the 5p rows out, the return at the top)	default 1.00e-16, at the top 1.70e-16 m^3/s; {'jung': ('kannari', {'k76': 'jung'}), 'kannari': ('kannari', {}), 'jung_no5p': ('kannari', {'k76': 'jung', 'no5...	pass
12	K.kr_fill	the Kr fill: Kr 0.995, F2 0.005, the cations Kr+ and Kr2+, the beam's pairs shared by the stopping powers, 0.28 Kr* per pair made in krypton, its energy per pair	[('Kr', 0.995), ('F2', 0.005)], cations ('Kr+', 'Kr2+'), pairs {'Kr+': 0.9978230972792193, 'F2+': 0.0021769027207806492}, Kr* per pair {'Kr*': 0.279390467238...	pass
13	K.energy_rows_Kr2_1e+16	the ladder's net energy from the electrons = the sum over the solver's processes of the ladder's events x the energy each takes, and every electron row on an excited species is a process of the solver (the Kr fill, the state without excited atoms, kinetic rates, the measured K76)	p_net/P 34.5379 vs 34.5379, rows not carried []	pass

	check	what	value	
14	C.jung	the scan 'jung': 75 cells x 13 frequencies (errors only at the Kr fill's 0.5 atm 1e14 cell), every point closed the same way for the electrons and the ions (residual against the point's largest particle rate below the loop's tolerance), the solver's net power to the excited gases against the ladder's, the residuals, the rounds, the boundary and floor points	75 cells, 971 points, 4 errors at ['Kr 0.5 atm 1e14'], closure off at 0 [], closure residual 2.2e-04 held / 1.2e-04 growing, energy 8.2e-03, solver 3.6e-05, ...	pass
15	C.kannari	the scan 'kannari': 75 cells x 13 frequencies (errors only at the Kr fill's 0.5 atm 1e14 cell), every point closed the same way for the electrons and the ions (residual against the point's largest particle rate below the loop's tolerance), the solver's net power to the excited gases against the ladder's, the residuals, the rounds, the boundary and floor points	75 cells, 967 points, 8 errors at ['Kr 0.5 atm 1e14'], closure off at 0 [], closure residual 2.5e-04 held / 1.3e-04 growing, energy 5.9e-03, solver 3.6e-05, ...	pass
16	C.jung_no5p	the scan 'jung_no5p': 75 cells x 13 frequencies (errors only at the Kr fill's 0.5 atm 1e14 cell), every point closed the same way for the electrons and the ions (residual against the point's largest particle rate below the loop's tolerance), the solver's net power to the excited gases against the ladder's, the residuals, the rounds, the boundary and floor points	75 cells, 975 points, 0 errors at [], closure off at 0 [], closure residual 2.6e-04 held / 1.6e-04 growing, energy 4.7e-03, solver 3.4e-05, ladder 6.6e-16, |...	pass
17	C.jung_kpshi	the scan 'jung_kpshi': 75 cells x 13 frequencies (errors only at the Kr fill's 0.5 atm 1e14 cell), every point closed the same way for the electrons and the ions (residual against the point's largest particle rate below the loop's tolerance), the solver's net power to the excited gases against the ladder's, the residuals, the rounds, the boundary and floor points	75 cells, 974 points, 1 errors at ['Kr 0.5 atm 1e14'], closure off at 0 [], closure residual 2.5e-04 held / 1.3e-04 growing, energy 6.3e-03, solver 3.6e-05, ...	pass
18	C.recompute	the Kr fill, 1 atm, 1e15, first frequency, solved again in a fresh process (the measured K76)	<eps> 2.842079513 vs 2.842079513, x_ Kr* 5.129746938e-05 vs 5.129746938e-05	pass
19	C.same_rates	there, the solver's rate of every excited process = the same integral over its distribution, and the excited atoms' rates (the round before) within the loop's tolerance	solver vs the integral 2.2e-16, excited atoms vs solver 4.1e-03	pass
20	B.complete	a best frequency at every cell but the incomplete one	75 cells, incomplete ['Kr 0.5 atm 1e14']	pass
21	B.recompute	the Kr fill, 1 atm, 1e15 analysed again from the scan (f_best, T_e, the marks)	8.64033e+12 vs 8.64033e+12 Hz, T_e 1.5604	pass

	check	what	value	
22	`L.solvers`	the two kinetic solvers at the seven stored states: mean energy, the wave's absorbed power and the net frequency, second over first, and the ions' share at 0.1 atm, 1e16, 95 GHz	7 states, |mean - 1| 1.6e-03, |power - 1| 1.6e-03, |nu - 1| 5.8e-03, shares 0.1928 / 0.1719, power there 1.00019, quant_IB residual 1.3e-04	pass
23	`L.recompute`	quant_IB's side at 1 atm, 1e15, 3 kV/cm, 1 THz solved again in this process (the scans' electron-ion model)	<eps> 2.5932833 vs stored 2.5932833	pass
24	`L.optics`	the optics on the second solver's distributions at the seven states against quant_IB's (alpha, R), the power its mobility implies against its own, and quant_IB's alpha at one state recomputed here	|alpha - 1| 1.6e-03, |R - 1| 3.4e-03, |P - 1| 1.3e-04; alpha 551.245 vs stored 551.245	pass
25	`L.cold`	the two full models from the same cold start (Kr fill, 1 atm, 1e15 seed, 38.9 kV/cm at 10 THz, no beam): the decay rate of n_e (the second solver 10 to 100 ns, the march 40 to 100 ns), T_e at 100 ns, the F2 burnt	decay 2.1695e+08 / 2.1496e+08 /s, T_e 1.7469 / 1.7416 eV, F2 burnt 0.0086 / 0.0144, T_g 303.4 K	pass
26	`P.Kr15`	the pulse Kr15: quasineutral (n_e = cations - F-), F2 never rising, time increasing to 1 us, the renewal rule (at most dt or a twentieth of the time elapsed), the wave's absorbed power at the target at every renewal, the closure, the checkpoints' 13-frequency scans	86 renewals, quasineutral to 0.0e+00, F2 monotone True, t monotone True, rule True, |P/P* - 1| 1.0e-04, closures ['source+loss'], checkpoints [0, 100, 300, 1...	pass
27	`P.Ar15`	the pulse Ar15: quasineutral (n_e = cations - F-), F2 never rising, time increasing to 1 us, the renewal rule (at most dt or a twentieth of the time elapsed), the wave's absorbed power at the target at every renewal, the closure, the checkpoints' 13-frequency scans	87 renewals, quasineutral to 0.0e+00, F2 monotone True, t monotone True, rule True, |P/P* - 1| 9.7e-05, closures ['source+loss'], checkpoints [0, 100, 300, 1...	pass
28	`P.Kr16`	the pulse Kr16: quasineutral (n_e = cations - F-), F2 never rising, time increasing to 1 us, the renewal rule (at most dt or a twentieth of the time elapsed), the wave's absorbed power at the target at every renewal, the closure, the checkpoints' 13-frequency scans	103 renewals, quasineutral to 0.0e+00, F2 monotone True, t monotone True, rule True, |P/P* - 1| 9.6e-05, closures ['source+loss'], checkpoints [0, 30, 100, 3...	pass

	check	what	value	
29	N.cells	the reference's cells: three fills x 5 pressures x 5 densities, those complete, those not	75 cells, 74 complete, incomplete ['Kr 0.5 atm 1e14']	pass
30	N.fills	per fill: T_e at f_best, the marks, the net energy the excited atoms take over P	{'Ne': ('T_e 1.396-4.694', {'': 5, 'a': 15, 'b': 5, 'ab': 0}, 'p_net/P 0.007-2.9'), 'Ar': ('T_e 0.874-2.646', {'': 8, 'a': 8, 'b': 9, 'ab': 0}, 'p_net/P 0.01...	pass
31	N.kr_over_ar	the Kr fill's best frequency over the Ar fill's at the same p and n_e	24 cells, 1.0394 (at Kr 0.2 atm 3e14) to 1.2013 (at Kr 2 atm 3e15)	pass
32	N.kr_over_ar_mech	the effective momentum-transfer frequency (omega mu_r/mu_i at f_best) of the Kr fill over the Ar fill's at the same cell, and how far its square root lies from the ratio of the best frequencies	nu ratio 1.081 to 1.521, sqrt vs f ratio within 0.035	pass
33	N.sigma_m	the momentum-transfer cross section of krypton over argon's at 1 to 3 eV	0.54 to 1.55	pass
34	N.band_kannari	the best frequency with the case 'kannari' against the reference, by fill (the cells complete in both)	{'Ne': (25, '0.9999-1.0099', 'Ne 1 atm 3e14'), 'Ar': (25, '0.9999-1.0467', 'Ar 2 atm 1e16'), 'Kr': (24, '0.9999-1.0491', 'Kr 2 atm 1e16')}	pass
35	N.band_jung_no5p	the best frequency with the case 'jung_no5p' against the reference, by fill (the cells complete in both)	{'Ne': (25, '0.9995-1.0541', 'Ne 1 atm 1e15'), 'Ar': (25, '1.0035-1.4596', 'Ar 2 atm 1e16'), 'Kr': (24, '1.0056-1.4362', 'Kr 2 atm 1e16')}	pass
36	N.band_jung_kpshi	the best frequency with the case 'jung_kpshi' against the reference, by fill (the cells complete in both)	{'Ne': (25, '0.9990-1.0000', 'Ne 2 atm 1e16'), 'Ar': (25, '0.9509-0.9999', 'Ar 1 atm 1e16'), 'Kr': (24, '0.8685-0.9998', 'Kr 0.5 atm 1e16')}	pass
37	N.step_band	stepwise ionization at the template (2.1 to 2.3 x Kannari's fit) or at none against Kannari's fit, in the calculation with his 5s → 5p fit (Ne and Ar, 50 cells each): the largest move of f_best	0.0144	pass
38	N.jung	Jung et al.'s lumped 5s → 5p cross section over Kannari's fit on Maxwellians at the reference's lowest and highest T_e	1.358 at 0.874 eV, 0.920 at 4.694 eV	pass

	check	what	value	
39	N.nist	the atomic levels and ionization limits of the ledger against the NIST database	{'Kr*': 0.0002, 'Kr**': 0.0035, 'Kr+': 0.0036, 'Ar*': 0.0004, 'Ar+': 0.0004, 'Ne*': 0.0001, 'Ne+': 0.0045}	pass
40	N.prad	the 5p manifold's lifetime: the 2J+1-weighted mean decay rate of the ten level lifetimes against the row's 29.0 ns	29.04 ns from 23.1 to 38.1 ns, the row 29.0 ns	pass
41	N.loop	the reference scan's points, the loop's rounds, the growth rate over the momentum-transfer frequency at the growing points, the points accepted at a flipping boundary and at the pulse floor	971 points, 3 to 72 rounds, g/nu up to 2.83e-03 (394 growing), boundary 5, floor 0	pass
42	N.return_5p	how the 5p manifold returns to 5s at the Kr fill, 1 atm, 1e15, 10 THz (the events)	{'collisions': '0.838', 'superelastic': '0.150', 'radiation': '0.012'}	pass
43	N.f2_burn	the time attachment alone takes the F2 down e-fold at f_best, and the slowest excited species' relaxation time	F2: 24.4 to 56.2 ns at 1e16, 436 to 974 ns at 1e15; excited atoms within 212 ns	pass
44	N.pulse_Kr15	the pulse Kr15: F2 below 1 % of its fill, n_e steady (|g| < 1e6/s), the rise of n_e, the fall of T_e, the field, the best frequency of the frozen state at the checkpoints	F2 < 1 % at 365 ns (half at 134.0), g ~ 0 at 443 ns, n_e x3.75, T_e 1.560 → 0.535, E 38.9 → 47.6 kV/cm, f_best 0 ns 8.64, 100 ns 8.80, 300 ns 7.18, 1000 ns...	pass
45	N.pulse_Ar15	the pulse Ar15: F2 below 1 % of its fill, n_e steady (|g| < 1e6/s), the rise of n_e, the fall of T_e, the field, the best frequency of the frozen state at the checkpoints	F2 < 1 % at 383 ns (half at 134.0), g ~ 0 at 465 ns, n_e x3.95, T_e 1.519 → 0.542, E 34.6 → 37.0 kV/cm, f_best 0 ns 7.48, 100 ns 7.71, 300 ns 6.89, 1000 ns...	pass
46	N.pulse_Kr16	the pulse Kr16: F2 below 1 % of its fill, n_e steady (|g| < 1e6/s), the rise of n_e, the fall of T_e, the field, the best frequency of the frozen state at the checkpoints	F2 < 1 % at 35 ns (half at 11.0), g ~ 0 at 92 ns, n_e x6.83, T_e 0.943 → 0.532, E 33.6 → 12.1 kV/cm, f_best 0 ns 19.19, 30 ns 29.11, 100 ns 52.86, 300 ns 5...	pass
47	N.recorded	the note's numbers computed from the scans, the best frequencies and the pulses	13 of 13 groups recorded	pass
48	N.inc_cell	the incomplete cell's frequencies closed in the reference and those accepted at the flipping boundary, and the cell with the most rounds	{'records': 9, 'boundary': 4}, most rounds at Kr 0.5 atm 1e14 (72)	pass

	check	what	value	
49	E.third	claim: the binding of Kr* over Kr's ionization energy is under a third (0.292)		pass
50	E.table75	claim: three fills, 75 points, the krypton fill with 0.5 % F2 ([('Kr', 0.995), ('F2', 0.005)])		pass
51	E.ans_kr_lo	printed 1.04: the Kr fill's f_best over the Ar fill's, smallest	1.03944	pass
52	E.ans_kr_hi	printed 1.2: the same, largest	1.20134	pass
53	E.ans_f01_lo	printed 1.05: the Kr fill's f_best at 0.1 atm, lowest [THz]	1.04983	pass
54	E.ans_f01_hi	printed 8.8: the same, highest	8.81888	pass
55	E.ans_f2_lo	printed 3.8: at 2 atm, lowest	3.75663	pass
56	E.ans_f2_hi	printed 26.9: at 2 atm, highest	26.8752	pass
57	E.ans_te_lo	printed 0.94: the Kr fill's T_e, lowest	0.940115	pass
58	E.ans_te_hi	printed 2.44: the same, highest	2.43674	pass
59	E.ans_b11	claim: the beam holds 11 of the Kr fill's cells (b marks {'': 7, 'a': 6, 'b': 11, 'ab': 0})		pass
60	E.ans_ne_lo	printed 0.933: the Ne fill over the calculation without excited atoms, smallest	0.932873	pass
61	E.ans_ne_hi	printed 0.997: the same, largest	0.996997	pass
62	E.ans_ar_lo	printed 0.646: the Ar fill, smallest	0.646198	pass
63	E.ans_ar_hi	printed 0.995: the Ar fill, largest	0.995163	pass
64	E.ans_fit	printed 5: Kannari's fit in place of the measurement raises f_best by up to (%)	4.91307	pass
65	E.ans_no5p	printed 1.46: the 5p rung's electron rows left out, up to	1.45957	pass
66	E.ans_kps	printed 13: the return at the measured bound lowers f_best by up to (%)	13.1509	pass
67	E.ans_kps_te	printed 19: and T_e there by (%)	19.0497	pass
68	E.ans_kps_where	claim: the most at Kr 0.5 atm 1e16 (a dense krypton point)		pass
69	E.ans_hold15	claim: at 1e15 the frozen state's f_best at 100 ns within 3 % of the start (1.0189, 1.0306)		pass
70	E.ans_f2_15lo	printed 0.36: F2 below 1 % of its fill at 1e15, earliest [us]	0.364652	pass
71	E.ans_f2_15hi	printed 0.38: the same, latest	0.382884	pass
72	E.ans_f2_16	printed 35: at 1e16 [ns]	34.7287	pass
73	E.ans_ne_rise_lo	printed 3.7: n_e runs up, least	3.74961	pass
74	E.ans_ne_rise_hi	printed 6.8: the same, most	6.83088	pass
75	E.ans_te_cool_lo	printed 0.53: the electrons cool to, lowest [eV]	0.531929	pass
76	E.ans_te_cool_hi	printed 0.54: the same, highest	0.541683	pass
77	E.ans_fmove_kr	printed -18: f_best at 1 us over the start, the Kr fill at 1e15 (%)	-17.9178	pass
78	E.ans_fmove_ar	printed -6: the Ar fill (%)	-6.09407	pass
79	E.ans_fmove_16	printed 2.8: the Kr fill at 1e16, times	2.75057	pass
80	E.ans_seven	claim: seven states from 1e12 to 1e16 (7, (1000000000000.0, 1e+16))		pass
81	E.ans_02	claim: within 0.2 % in mean energy (1.65e-03) and power (1.64e-03)		pass

	check	what	value	
82	E.ans_cold	claim: the two full models' decay rates 1 % apart (2.1695e+08 vs 2.1496e+08) and T_e 0.2 % apart (1.7469 vs 1.7416)		pass
83	E.fills	claim: the fills' mole fractions {'Ne': [('Ne', 0.948), ('Kr', 0.05), ('F2', 0.002)], 'Ar': [('Ar', 0.597), ('Kr', 0.4), ('F2', 0.003)], 'Kr': [('Kr', 0.995), ('F2', 0.005)]}		pass
84	E.grid	claim: pressures (0.1, 0.2, 0.5, 1.0, 2.0), densities (100000000000000.0, 300000000000000.0, 1000000000000000.0, 3000000000000000.0, 1e+16)		pass
85	E.freqs	claim: 13 frequencies, five per decade, 0.1 to 25.12 THz		pass
86	E.rmax	claim: R_max 0.05		pass
87	E.cations	claim: the cations {'Ne': ('Ne+', 'Ne2+', 'NeKr+', 'Kr+', 'Kr2+'), 'Ar': ('Ar+', 'Ar2+', 'ArKr+', 'Kr+', 'Kr2+'), 'Kr': ('Kr+', 'Kr2+')}		pass
88	E.epssec	printed 4.506: the mean energy of the beam's slow secondaries	4.506	pass
89	E.tau	claim: the pulse 1e-06 s		pass
90	E.kr915	printed 9.915: Kr* on the ledger	9.915	pass
91	E.kr1130	printed 11.3: Kr** on the ledger	11.3	pass
92	E.ar11548	printed 11.548: Ar* on the ledger	11.548	pass
93	E.ne16619	printed 16.619: Ne* on the ledger	16.619	pass
94	E.kr_species	claim: the Kr fill carries Kr*, Kr** and Kr2*		pass
95	E.nist	claim: the atoms' levels within 0.005 eV of NIST (largest 0.0045 eV)		pass
96	E.bethe	claim: the stopping share at 600 keV (6.0e5 eV in the code)		pass
97	E.beam	claim: the beam's energies per pair and excited atoms per pair as the calculation uses them		pass
98	E.harp	printed 8.47e-10: Kr* + F2, all channels, cm3/s	8.47e-10	pass
99	E.yield	printed 0.9: the KrF* yield of Kr* + F2	0.9	pass
100	E.jung_meas	claim: Jung's tables: 2 to 300 eV, +-35 % (1s5), +-40 % (1s3), the ten 5p levels		pass
101	E.jung_weights	claim: the 5s manifold lumped 5:3:1:3 (weights {'1s5': 0.6666666666666666, '1s3': 0.3333333333333333})		pass
102	E.gap	printed 1.385: the ledger's gap Kr* → Kr** [eV]	1.385	pass
103	E.jung_lo	printed 1.36: the measured lumped rate over Kannari's fit at the table's coldest T_e	1.35791	pass
104	E.jung_tlo	printed 0.87: that T_e [eV]	0.873805	pass
105	E.jung_hi	printed 0.92: the same at the hottest	0.919652	pass
106	E.jung_thi	printed 4.69: that T_e [eV]	4.69366	pass
107	E.jung_cal	claim: both ratios within the measurement's 35 to 40 %		pass

	check	what	value	
108	E.tau5p_lo	printed 23.1: the ten 5p lifetimes, shortest [ns]	23.1	pass
109	E.tau5p_hi	printed 38.1: the same, longest	38.1	pass
110	E.tau5p	printed 29.0: the 2J+1 mean [ns]	29.0377	pass
111	E.gap_light	printed 1.385: the energy that leaves as light	1.385	pass
112	E.kps	printed 1e-10: the return by collisions, the default [cm3/s]	1e-10	pass
113	E.gazeli	claim: Gazeli's measured total loss, the band's top: 1.70e-10 cm3/s		pass
114	E.ret_coll	printed 84: the 5p manifold's return by collisions at Kr 1 atm 1e15 10 THz (%)	83.7806	pass
115	E.ret_super	printed 15: superelastically (%)	15.0325	pass
116	E.ret_rad	printed 1.2: by radiation (%)	1.18689	pass
117	E.kstepA	printed 7e-08: Kannari's stepwise fit, A [cm3/s]	7e-08	pass
118	E.kstepB	printed 0.68: its exponent B	0.68	pass
119	E.kstepE	printed 4.04: its activation E [eV]	4.04	pass
120	E.jh49	claim: Johnson & Hunter's row 49, 4.8e-8 from Vriens		pass
121	E.tmpl_lo	printed 2.1: the template over Kannari's fit at the table's temperatures, smallest	2.09613	pass
122	E.tmpl_hi	printed 2.3: the same, largest	2.30965	pass
123	E.step_band	claim: the stepwise band moved f_best by less than 1.5 % (largest 0.0144)		pass
124	E.box	printed 0.2: the width of a Penning electron's line	0.2	pass
125	E.pen_lo	printed 2.1: the lowest Penning release energy	2.104	pass
126	E.pen_hi	printed 12.0: the highest	12	pass
127	E.gnu	claim: g/nu_m below 3e-3 at every growing point (largest 2.830e-03)		pass
128	E.tol	claim: the loop's tolerance 0.002		pass
129	E.floor	claim: the pulse floor 1000000.0 /s		pass
130	E.bnd_rule	claim: the boundary rule: within 10.0 tolerances, three flips		pass
131	E.rounds	claim: the loop took 3 to 72 rounds, the most at Kr 0.5 atm 1e14		pass
132	E.inc_cell	claim: the incomplete cell: {'records': 9, 'boundary': 4}		pass
133	E.kr_lo	printed 1.04: the Kr fill over the Ar fill, smallest	1.03944	pass
134	E.kr_hi	printed 1.2: the same, largest	1.20134	pass
135	E.kr_where	claim: the least at Kr 0.2 atm 3e14, the most at Kr 2 atm 3e15		pass
136	E.nu_lo	printed 1.08: nu_m of the Kr fill over the Ar fill's at f_best, smallest	1.08116	pass
137	E.nu_hi	printed 1.52: the same, largest	1.52106	pass
138	E.nu_sqrt	printed 3.5: sqrt of the nu ratio against the f ratio, largest difference (%)	3.46553	pass
139	E.sm_lo	printed 0.54: sigma_m(Kr)/sigma_m(Ar) at 1 eV	0.536335	pass

	check	what	value	
140	E.sm_hi	printed 1.55: the same at 3 eV	1.54808	pass
141	E.ar40	claim: the argon fill is 40 % krypton		pass
142	E.te_kr_lo	printed 0.94: the Kr fill's T_e, lowest	0.940115	pass
143	E.te_kr_hi	printed 2.44: highest	2.43674	pass
144	E.te_ar_lo	printed 0.87: the Ar fill's T_e, lowest	0.873805	pass
145	E.te_ar_hi	printed 2.65: highest	2.64608	pass
146	E.te_ne_lo	printed 1.4: the Ne fill's T_e, lowest	1.39604	pass
147	E.te_ne_hi	printed 4.69: highest	4.69366	pass
148	E.flags_a	claim: a marks [15, 8, 6]		pass
149	E.flags_b	claim: b marks [5, 9, 11]		pass
150	E.pnet_ne_lo	printed 0.007: p_net/P in the Ne fill, smallest	0.0070093	pass
151	E.pnet_ne_hi	printed 2.9: largest	2.89352	pass
152	E.pnet_ar_lo	printed 0.012: the Ar fill, smallest	0.0119025	pass
153	E.pnet_ar_hi	printed 15.0: largest	15.0481	pass
154	E.pnet_kr_lo	printed 0.017: the Kr fill, smallest	0.0173269	pass
155	E.pnet_kr_hi	printed 23.5: largest	23.4638	pass
156	E.pnet_where	claim: the most at 2 atm 1e16 in each: ['Ne 2 atm 1e16', 'Ar 2 atm 1e16', 'Kr 2 atm 1e16']		pass
157	E.vs_ne_lo	printed 0.933: the Ne fill over the calculation without excited atoms, smallest	0.932873	pass
158	E.vs_ne_hi	printed 0.997: largest	0.996997	pass
159	E.vs_ar_lo	printed 0.646: the Ar fill, smallest	0.646198	pass
160	E.vs_ar_hi	printed 0.995: largest	0.995163	pass
161	E.falls	claim: falling at every point (25 + 25)		pass
162	E.vs_where	claim: the most at Ar 2 atm 1e16		pass
163	E.f0	printed 34.9: f_best there without the excited atoms [THz]	34.8906	pass
164	E.f1	printed 22.5: and with them	22.5463	pass
165	E.tev_ar_lo	printed 0.55: T_e over without, the Ar fill, smallest	0.546861	pass
166	E.tev_ar_hi	printed 0.99: largest	0.990902	pass
167	E.tev_ne_lo	printed 0.83: the Ne fill, smallest	0.831089	pass
168	E.tev_ne_hi	printed 0.99: largest	0.992994	pass
169	E.bcells	claim: every band at 74 cells		pass
170	E.bk_lo	printed 1.0: Kannari's fit over the reference, smallest	0.999887	pass
171	E.bk_hi	printed 1.049: largest	1.04913	pass
172	E.bk_ne	printed 1.01: the Ne fill, largest	1.00989	pass
173	E.bk_ar	printed 1.047: the Ar fill, largest	1.04674	pass
174	E.bk_kr	printed 1.049: the Kr fill, largest	1.04913	pass
175	E.bk_where	claim: the most at Kr 2 atm 1e16		pass
176	E.bk_f0	printed 26.9: f_best there, the reference [THz]	26.8752	pass
177	E.bk_f1	printed 28.2: with the fit	28.1956	pass
178	E.bk_te	printed 6.7: the electrons hotter by at most (%)	6.67175	pass
179	E.b5p_lo	printed 1.0: the 5p rows out over the reference, smallest	0.999536	pass
180	E.b5p_hi	printed 1.46: largest	1.45957	pass
181	E.b5p_ne	printed 1.054: the Ne fill	1.05409	pass

	check	what	value	
182	E.b5p_ar	printed 1.46: the Ar fill	1.45957	pass
183	E.b5p_kr	printed 1.436: the Kr fill	1.43624	pass
184	E.b5p_where	claim: the most at Ar 2 atm 1e16		pass
185	E.b5p_f	printed 32.9: f_best there with the 5p rows out [THz]	32.9078	pass
186	E.br_lo	printed 0.868: the return at the measured bound over the reference, smallest	0.868491	pass
187	E.br_hi	printed 1.0: largest	0.999997	pass
188	E.br_ne	printed 0.999: the Ne fill, smallest	0.999023	pass
189	E.br_ar	printed 0.951: the Ar fill, smallest	0.950916	pass
190	E.br_kr	printed 0.868: the Kr fill, smallest	0.868491	pass
191	E.br_where	claim: the most at Kr 0.5 atm 1e16		pass
192	E.br_f0	printed 15.6: f_best there, the reference [THz]	15.5601	pass
193	E.br_f1	printed 13.5: at the bound	13.5138	pass
194	E.br_te	printed 0.81: T_e there over the reference	0.809503	pass
195	E.br_13	printed 13: the return rate's uncertainty in f_best (%)	13.1509	pass
196	E.br_19	printed 19: in T_e (%)	19.0497	pass
197	E.arspread	printed 0.28: the Ar levels above the lumped Ar*, largest (eV)	0.28	pass
198	E.gnu2	claim: the same g/nu_m bound		pass
199	E.nbnd	claim: one boundary point of the Ar fill ([('Ar 0.5 atm 3e14', 630957000000.0)])		pass
200	E.settle	claim: the excited atoms settle within 220 ns (slowest 211.5 ns)		pass
201	E.tf2_16lo	printed 24: attachment's F2 e-folding at 1e16, shortest [ns]	24.4416	pass
202	E.tf2_16hi	printed 56: longest	56.1545	pass
203	E.tf2_15lo	printed 0.44: at 1e15, shortest [us]	0.435913	pass
204	E.tf2_15hi	printed 0.97: longest	0.973668	pass
205	E.march_rule	claim: the drift 0.02, the renewal floor in the code		pass
206	E.march_held	claim: T_g 300 K held, the excimers sinks		pass
207	E.pulse_points	claim: the three pulses' points and frequencies		pass
208	E.pulse_f	claim: the wave's frequencies [10000000000000.0, 7700000000000.0, 19150000000000.0] and the points		pass
209	E.burn_lo	printed 300000000.0: F2 molecules burnt per electron per second at t = 0, least	3.18072e+08	pass
210	E.burn_hi	printed 600000000.0: most	6.43812e+08	pass
211	E.half15	printed 134: half the F2 gone at 1e15 [ns] (both fills)	134.01	pass
212	E.half15_ar	claim: the same in the Ar fill		pass
213	E.gone_kr	printed 365: F2 below 1 %, the Kr fill [ns]	364.652	pass
214	E.gone_ar	printed 383: the Ar fill [ns]	382.884	pass
215	E.rise_kr	printed 3.75: n_e runs up, the Kr fill	3.74961	pass
216	E.rise_ar	printed 3.95: the Ar fill	3.95174	pass
217	E.gzero_lo	printed 0.44: n_e steady (|g| < 1e6/s) by, earliest [us]	0.443237	pass
218	E.gzero_hi	printed 0.47: latest	0.465398	pass

	check	what	value	
219	E.te0_kr	printed 1.56: T_e at t = 0, the Kr fill	1.5596	pass
220	E.te0_ar	printed 1.52: the Ar fill	1.51863	pass
221	E.te1_kr	printed 0.535: at 1 us, the Kr fill	0.53494	pass
222	E.te1_ar	printed 0.542: the Ar fill	0.541683	pass
223	E.xkr_kr	printed 10: Kr* accumulates, the Kr fill (times)	10.3634	pass
224	E.xkr_ar	printed 8: the Ar fill	8.08276	pass
225	E.E0_kr	printed 38.9: the field at t = 0, the Kr fill [kV/cm]	38.8871	pass
226	E.E1_kr	printed 47.6: at 1 us	47.6354	pass
227	E.f100_kr	printed 2: f_best at 100 ns over the start, the Kr fill (%)	1.88791	pass
228	E.f100_ar	printed 3: the Ar fill (%)	3.05967	pass
229	E.fend_kr	printed 0.82: at 1 us over the start, the Kr fill	0.820822	pass
230	E.fend_ar	printed 0.94: the Ar fill	0.939059	pass
231	E.ne100_kr	printed 25: n_e at 100 ns over the start, the Kr fill (%)	25.0987	pass
232	E.ne100_ar	printed 28: the Ar fill (%)	27.7966	pass
233	E.te100_kr	printed 1.37: T_e at 100 ns, the Kr fill	1.37008	pass
234	E.te100_ar	printed 1.34: the Ar fill	1.33598	pass
235	E.f2100_kr	printed 0.61: F2 at 100 ns over its fill, the Kr fill	0.612692	pass
236	E.f2100_ar	printed 0.6: the Ar fill	0.600919	pass
237	E.hold3	claim: within 3 % in f_best at 100 ns (1.0189, 1.0306)		pass
238	E.half16	printed 11: half the F2 gone at 1e16 [ns]	11	pass
239	E.gone16	printed 35: below 1 % [ns]	34.7287	pass
240	E.rise16	printed 6.8: n_e runs up (times)	6.83088	pass
241	E.gzero16	printed 92: n_e steady by [ns]	92.1457	pass
242	E.te0_16	printed 0.94: T_e at t = 0	0.942549	pass
243	E.te1_16	printed 0.53: at 1 us	0.531929	pass
244	E.E0_16	printed 33.6: the field at t = 0 [kV/cm]	33.5685	pass
245	E.E1_16	printed 12.1: at 1 us	12.0529	pass
246	E.a0_16	printed 10.0: alpha at t = 0 [1/m]	10.0409	pass
247	E.a1_16	printed 78: at 1 us	78.3845	pass
248	E.depth16	printed 1.3: 1/alpha at 1 us [cm]	1.27576	pass
249	E.f30_16	printed 1.52: f_best at 30 ns over the start	1.51723	pass
250	E.fend_16	printed 2.75: at 1 us over the start	2.75057	pass
251	E.fend_16_thz	printed 53: f_best at 1 us [THz]	52.7725	pass
252	E.f100_16	claim: from 100 ns on: 100, 300 and 1000 ns within 0.3 % of each other ({'0': 19186025949982.11, '30': 29109668959022.47, '100': 52856943530711.71, '300': 52825798062356.71, '1000': 52772522820037.086})		pass
253	E.ne10_16	printed 32: n_e at 10 ns over the start (%)	31.6008	pass
254	E.f210_16	printed 0.54: F2 at 10 ns over its fill	0.53612	pass

	check	what	value	
255	E.sv_range	claim: seven states: n_e (1000000000000.0, 1e+16), p (0.1, 1.0), f (95000000000.0, 10000000000000.0), ions' share up to 0.250		pass
256	E.sv_agree	claim: mean energy within 0.2 % (1.65e-03), power within 0.2 % (1.64e-03), nu within 0.6 % (5.77e-03)		pass
257	E.sv_share1	printed 0.19: the ions' share at 0.1 atm 1e16 95 GHz, first solver	0.192773	pass
258	E.sv_share2	printed 0.17: second solver	0.171885	pass
259	E.sv_p95	printed 0.02: the total power there, |ratio - 1| (%)	0.0191881	pass
260	E.opt_agree	claim: alpha within 0.2 % (1.59e-03), R within 0.3 % (3.37e-03), the mobility's power within 1e-4 of the second solver's own (1.35e-04)		pass
261	E.opt_a_lo	printed 2.3: alpha over the seven states, smallest [1/m]	2.26871	pass
262	E.opt_a_hi	printed 34000.0: largest	33681.1	pass
263	E.opt_r_lo	printed 3e-08: R, smallest	2.50963e-08	pass
264	E.opt_r_hi	printed 0.91: largest	0.907518	pass
265	E.opt_p	claim: the mobility's power = the second solver's own to 1e-4 (1.35e-04)		pass
266	E.cold_state	claim: the cold start: the march's field 38900.0 V/cm, its closure ['growth'], both run 100 ns, no beam		pass
267	E.cold_g2	printed 217000000.0: the second model's decay rate, 10 to 100 ns [1/s]	2.16952e+08	pass
268	E.cold_g1	printed 215000000.0: the march's, 40 to 100 ns	2.14962e+08	pass
269	E.cold_g_apart	printed 1: the two apart (%)	0.925893	pass
270	E.cold_te2	printed 1.747: T_e at 100 ns, the second model	1.74692	pass
271	E.cold_te1	printed 1.742: the march	1.74164	pass
272	E.cold_te_apart	printed 0.3: apart (%)	0.302988	pass
273	E.cold_decades	claim: the second model loses three decades of n_e in its first 10 ns (3.29)		pass
274	E.cold_f2_2	printed 0.9: F2 burnt in 100 ns, the second model (%)	0.861654	pass
275	E.cold_f2_1	printed 1.4: the march (%)	1.43963	pass
276	E.cold_tg	printed 3.4: the gas warms by [K]	3.41622	pass
277	E.st_13	printed 13: the return rate's uncertainty in f_best (%)	13.1509	pass
278	E.st_19	printed 19: in T_e (%)	19.0497	pass
279	E.st_hold	claim: the first 100 ns at 1e15 and 10 ns at 1e16		pass
280	E.st_02	claim: the solvers to 0.2 %		pass
281	S.semicolon	no semicolon in the prose	0 found	pass
282	S.emdash	no em-dash in the prose	0 found	pass
283	S.placeholders	no unfilled placeholder	0 found []	pass

	check	what	value	
284	`S.self_contained`	no local file, path, model or unpublished note named in the printed source, the generated tables or the PDF's text	source [], PDF []	pass

B The rows

The heavy-particle reactions of the excited atoms, Table 9, at 300 K, and the reactions of the cations, Table 10, in SI units as the ion balance takes them, $\mathrm{m^3\,s^{-1}}$ for two bodies and $\mathrm{m^6\,s^{-1}}$ for three, with T_e in eV. F(a, s) is $aN/(N^2+1.7)\,s$ and L(A, b, c) is $AN/(N^2+bN+c)$, both in $\mathrm{cm^3\,s^{-1}}$ with N in amagat. A rate run as a band stands at its default.

Table 9: The heavy-particle reactions of the excited atoms.

reaction	rate	standing	source
$2\ \mathrm{Kr^*} \rightarrow \mathrm{Kr^+} + \mathrm{Kr} + e$	5.00×10^{-10} $\mathrm{cm^3\,s^{-1}}$	printed	5.0×10^{-10} of the second table, run as a band, [7]
$\mathrm{Kr^*} + \mathrm{F_2} \rightarrow \mathrm{KrF(B)} + \mathrm{F}$	4.73×10^{-10} $\mathrm{cm^3\,s^{-1}}$	measured	the mean of three measurements, $\mathrm{KrF^*}$ yield 0.90, B to C 62 to 38, [13, 14, 15]
$\mathrm{Kr^*} + \mathrm{F_2} \rightarrow \mathrm{KrF(C)} + \mathrm{F}$	2.90×10^{-10} $\mathrm{cm^3\,s^{-1}}$	measured	as the row above, the C share, [13, 14, 15]
$\mathrm{Kr^*} + \mathrm{F_2} \rightarrow \mathrm{Kr} + 2\ \mathrm{F}$	8.47×10^{-11} $\mathrm{cm^3\,s^{-1}}$	measured	the 10 % without $\mathrm{KrF^*}$, [13]
$\mathrm{Kr^*} + 2\ \mathrm{Kr} \rightarrow \mathrm{Kr_2^*} + \mathrm{Kr}$	1.70×10^{-32} $\mathrm{cm^6\,s^{-1}}$, $\times(T_g/300)^{-0.64}$	measured	the temperature law of the second, [14, 16]
$\mathrm{Kr_2^*} \rightarrow 2\ \mathrm{Kr}$	3.70×10^{6} $\mathrm{s^{-1}}$	measured	270 ns, [17]
$\mathrm{Kr_2^*} + \mathrm{F_2} \rightarrow \mathrm{KrF(B)} + \mathrm{Kr} + \mathrm{F}$	1.05×10^{-10} $\mathrm{cm^3\,s^{-1}}$	measured	the total measured, split evenly, [18]
$\mathrm{Kr_2^*} + \mathrm{F_2} \rightarrow \mathrm{Kr_2F^*} + \mathrm{F}$	1.05×10^{-10} $\mathrm{cm^3\,s^{-1}}$	measured	the total measured, split evenly, [18]
$\mathrm{Kr_2^*} + \mathrm{F} \rightarrow \mathrm{KrF(B)} + \mathrm{Kr}$	0 $\mathrm{cm^3\,s^{-1}}$	none, 0	no measurement
$2\ \mathrm{Kr_2^*} \rightarrow \mathrm{Kr_2^+} + 2\ \mathrm{Kr} + e$	0 $\mathrm{cm^3\,s^{-1}}$	none, 0	no measurement
$2\ \mathrm{Ar^*} \rightarrow \mathrm{Ar^+} + \mathrm{Ar} + e$	0 $\mathrm{cm^3\,s^{-1}}$	none, 0	no measurement
$\mathrm{Ar^*} + \mathrm{F_2} \rightarrow \mathrm{ArF(B)} + \mathrm{F}$	4.05×10^{-10} $\mathrm{cm^3\,s^{-1}}$	measured	the mean of three groups, $\mathrm{ArF^*}$ yield 0.5, [13, 21, 14, 22]
$\mathrm{Ar^*} + \mathrm{F_2} \rightarrow \mathrm{Ar} + 2\ \mathrm{F}$	4.05×10^{-10} $\mathrm{cm^3\,s^{-1}}$	measured	the half without $\mathrm{ArF^*}$, [22]
$\mathrm{Ar^*} + 2\ \mathrm{Ar} \rightarrow \mathrm{Ar_2^*} + \mathrm{Ar}$	6.00×10^{-33} $\mathrm{cm^6\,s^{-1}}$	measured	[14]
$\mathrm{Ar_2^*} \rightarrow 2\ \mathrm{Ar}$	3.12×10^{5} $\mathrm{s^{-1}}$	measured	the triplet, 3.2 μs, [20]
$\mathrm{Ar_2^*} + \mathrm{F_2} \rightarrow \mathrm{ArF(B)} + \mathrm{Ar} + \mathrm{F}$	4.73×10^{-10} $\mathrm{cm^3\,s^{-1}}$	measured	$\mathrm{ArF^*}$ share 0.91, [21, 22]
$\mathrm{Ar_2^*} + \mathrm{F_2} \rightarrow \mathrm{Ar_2F^*} + \mathrm{F}$	4.68×10^{-11} $\mathrm{cm^3\,s^{-1}}$	measured	the rest, $\mathrm{Ar_2F^*}$, [21, 22]
$\mathrm{Ar_2^*} + \mathrm{F} \rightarrow \mathrm{ArF(B)} + \mathrm{Ar}$	0 $\mathrm{cm^3\,s^{-1}}$	none, 0	no measurement
$2\ \mathrm{Ar_2^*} \rightarrow \mathrm{Ar_2^+} + 2\ \mathrm{Ar} + e$	0 $\mathrm{cm^3\,s^{-1}}$	none, 0	no measurement
$2\ \mathrm{Ne^*} \rightarrow \mathrm{Ne^+} + \mathrm{Ne} + e$	5.00×10^{-10} $\mathrm{cm^3\,s^{-1}}$	printed	[25]
$\mathrm{Ne^*} + 2\ \mathrm{Ne} \rightarrow \mathrm{Ne_2^*} + \mathrm{Ne}$	4.00×10^{-34} $\mathrm{cm^6\,s^{-1}}$, $\times(T_g/300)^{-0.75}$	printed	[24, 23]
$\mathrm{Ne^*} + \mathrm{Kr} \rightarrow \mathrm{Kr^+} + \mathrm{Ne} + e$	6.50×10^{-11} $\mathrm{cm^3\,s^{-1}}$	analog	of $\mathrm{Ne^*} + \mathrm{Ar}$, [23]
$\mathrm{Ne^*} + \mathrm{Kr} \rightarrow \mathrm{NeKr^+} + e$	2.30×10^{-11} $\mathrm{cm^3\,s^{-1}}$	analog	of $\mathrm{Ne^*} + \mathrm{Ar}$, associative, [23]
$\mathrm{Ne^*} + \mathrm{F_2} \rightarrow \mathrm{NeF^*} + \mathrm{F}$	5.30×10^{-10} $\mathrm{cm^3\,s^{-1}}$	printed	[23]
$\mathrm{Kr^*} + \mathrm{Kr} + \mathrm{Ne} \rightarrow \mathrm{Kr_2^*} + \mathrm{Ne}$	0 $\mathrm{cm^6\,s^{-1}}$, $\times(T_g/300)^{-0.75}$	none, 0	analog of $\mathrm{Ar^*} + \mathrm{Ar} + \mathrm{Ne}$, a band from 0
$\mathrm{Ne_2^*} \rightarrow 2\ \mathrm{Ne}$	7.50×10^{7} $\mathrm{s^{-1}}$	printed	[24]
$\mathrm{Ne_2^*} + \mathrm{Kr} \rightarrow \mathrm{Kr^+} + 2\ \mathrm{Ne} + e$	6.50×10^{-11} $\mathrm{cm^3\,s^{-1}}$	analog	of $\mathrm{Ne_2^*} + \mathrm{Ar}$, [23, 26]
$2\ \mathrm{Ne_2^*} \rightarrow \mathrm{Ne_2^+} + 2\ \mathrm{Ne} + e$	5.00×10^{-10} $\mathrm{cm^3\,s^{-1}}$	printed	[26]
$\mathrm{Ar^*} + \mathrm{Kr} \rightarrow \mathrm{Kr^*} + \mathrm{Ar}$	6.20×10^{-12} $\mathrm{cm^3\,s^{-1}}$	printed	[8]
$\mathrm{Kr^*} + \mathrm{Kr} + \mathrm{Ar} \rightarrow \mathrm{Kr_2^*} + \mathrm{Ar}$	2.70×10^{-32} $\mathrm{cm^6\,s^{-1}}$, $\times(T_g/300)^{-0.75}$	printed	[7]
$\mathrm{Kr^*} + 2\ \mathrm{Ar} \rightarrow \mathrm{ArKr^*} + \mathrm{Ar}$	1.00×10^{-32} $\mathrm{cm^6\,s^{-1}}$	printed	[7, 8]

reaction	rate	standing	source
$Ar^* + Kr + Ar \rightarrow ArKr^* + Ar$	1.00×10^{-32} cm^6 s^{-1}	printed	[7, 8]
$Ar_2^* + Kr \rightarrow Kr^* + 2\ Ar$	4.00×10^{-10} cm^3 s^{-1}	printed	[7, 8]
$ArKr^* + F_2 \rightarrow KrF(B) + Ar + F$	3.00×10^{-10} cm^3 s^{-1}	printed	split evenly to B and C, [7]
$ArKr^* + F_2 \rightarrow KrF(C) + Ar + F$	3.00×10^{-10} cm^3 s^{-1}	printed	split evenly to B and C, [7]
$ArKr^* + F_2 \rightarrow ArKrF^* + F$	3.00×10^{-10} cm^3 s^{-1}	printed	[7]
$ArKr^* + Kr \rightarrow Kr_2^* + Ar$	1.00×10^{-10} cm^3 s^{-1}	printed	[7, 8]
$ArKr^* \rightarrow Ar + Kr$	8.00×10^{7} s^{-1}	estimated	estimated there, [7]
$Kr^{**} + Ne \rightarrow Kr^* + Ne$	1.00×10^{-10} cm^3 s^{-1}	estimated	of $Kr^{**} + Ar$, estimated there, [7]
$Kr^{**} + Ar \rightarrow Kr^* + Ar$	1.00×10^{-10} cm^3 s^{-1}	estimated	estimated there, [7]
$Kr^{**} + Kr \rightarrow Kr^* + Kr$	1.00×10^{-10} cm^3 s^{-1}	estimated	of $Kr^{**} + Ar$, estimated there, run as a band to the measured total loss of one $5p'$ level, [7, 28]
$Kr^{**} + F_2 \rightarrow KrF(B) + F$	4.05×10^{-10} cm^3 s^{-1}	printed	split evenly to B and C, [8]
$Kr^{**} + F_2 \rightarrow KrF(C) + F$	4.05×10^{-10} cm^3 s^{-1}	printed	split evenly to B and C, [8]
$Kr^{**} \rightarrow Kr^*$	3.45×10^{7} s^{-1}	measured	the $2J+1$ mean of the ten level lifetimes, 29.0 ns, [29]

Table 10: The reactions of the cations.

reaction	rate	standing	source
$Ne^+ + Ne + Ne \rightarrow Ne_2^+ + Ne$	4.4×10^{-44}	printed	[25]
$Ne^+ + Kr \rightarrow Kr^+ + Ne$	1×10^{-17}	analog	of $Ne^+ + Ar$ and $Ne^+ + Xe$, [26]
$Ne^+ + Kr + Ne \rightarrow NeKr^+ + Ne$	1×10^{-43}	printed	assumed there, [27]
$Ne_2^+ + Kr \rightarrow Kr^+ + Ne + Ne$	5×10^{-19}	printed	the published upper bound, [27]
$e + Ne_2^+ \rightarrow Ne + Ne^*$	$3.7 \times 10^{-14}\, T_e^{-0.43}$	printed	exponent of the second, [25, 26]
$NeKr^+ + Kr \rightarrow Kr^+ + Ne + Kr$	1.1×10^{-15}	analog	of $NeAr^+ + Ar$, [23]
$NeKr^+ + Kr \rightarrow Kr_2^+ + Ne$	1.2×10^{-17}	analog	of $NeAr^+ + Ar$, [23]
$e + NeKr^+ \rightarrow Kr^{**} + Ne$	$1 \times 10^{-13}\, T_e^{-1/2}$	printed	[27]
$Kr^+ + Kr + Ne \rightarrow Kr_2^+ + Ne$	2.5×10^{-43}	estimated	within the measured analogs
$Kr^+ + Kr + Kr \rightarrow Kr_2^+ + Kr$	2.5×10^{-43}	printed	[8, 7, 27]
$e + Kr_2^+ \rightarrow Kr^* + Kr$	$2.24 \times 10^{-13}\, T_e^{-1/2}$	printed	both branches on $T_e^{-1/2}$, [7]
$Ar^+ + Ar + Ar \rightarrow Ar_2^+ + Ar$	2.5×10^{-43}	printed	[7]
$Ar^+ + Kr \rightarrow Kr^+ + Ar$	3×10^{-17}	printed	[7]
$Ar^+ + Kr + Ar \rightarrow ArKr^+ + Ar$	1×10^{-43}	printed	[7, 8]
$Ar_2^+ + Kr \rightarrow Kr^+ + Ar + Ar$	7.5×10^{-16}	printed	[7, 8]
$e + Ar_2^+ \rightarrow Ar^* + Ar$	$1.14 \times 10^{-13}\, T_e^{-1/2}$	printed	both branches on $T_e^{-1/2}$, [7]
$Kr^+ + Ar + Ar \rightarrow ArKr^+ + Ar$	1×10^{-43}	printed	[7, 8]
$ArKr^+ + Kr \rightarrow Kr_2^+ + Ar$	3.2×10^{-16}	printed	[7, 8]
$e + ArKr^+ \rightarrow Kr^{**} + Ar$	$1.93 \times 10^{-13}\, T_e^{-1/2}$	printed	on $T_e^{-1/2}$, [7]
$Kr^+ + Kr + Ar \rightarrow Kr_2^+ + Ar$	2.5×10^{-43}	printed	[7, 8]
$Kr^+ + F^- \rightarrow KrF^*$	F(2.4, 3.417×10^{-6})	form	this calculation's form for the turnover the source discusses in its Sec. II.B, [7]
$Kr_2^+ + F^- \rightarrow KrF^* + Kr$	F(2.65, 3.096×10^{-6})	form	this calculation's form for the turnover the source discusses in its Sec. II.B, [7]
$Ne^+ + F^- \rightarrow NeF^*$	L(2.8×10^{-5}, 1.44, 10.9)	printed	[23]
$Ne_2^+ + F^- \rightarrow NeF^* + Ne$	L(3.66×10^{-5}, 1.74, 17.3)	printed	[23]
$NeKr^+ + F^- \rightarrow KrF^* + Ne$	L(5.27×10^{-5}, 6.81, 20.3)	analog	of $NeAr^+ + F^-$, [23]
$Ar^+ + F^- \rightarrow ArF^*$	F(2.4, 3.417×10^{-6})	form	this calculation's form for the turnover the source discusses in its Sec. II.B, [7]
$Ar_2^+ + F^- \rightarrow ArF^* + Ar$	F(2.65, 3.096×10^{-6})	form	this calculation's form for the turnover the source discusses in its Sec. II.B, [7]

reaction	rate	standing	source
$ArKr^+ + F^- \to KrF^* + Ar$	$F(2.65,\ 3.096 \times 10^{-6})$	form	this calculation's form for the turnover the source discusses in its Sec. II.B, [7]

C Every point

Table 11: The 75 points at $L = 10$ cm and 300 kW/cm^3 with the excited atoms inside the electrons' balances and the measured 5s to 5p cross section: the best frequency in THz, the electron temperature at it in eV, the marks of Table 3, the mole fraction of Kr^* in units of 10^{-4}, the net energy the excited atoms take from the electrons over the absorbed power, then the best frequency of each band over this one, with Kannari's fit in place of the measured cross section (k), with the 5p rung's electron collisions left out ($5p$) and with the collisional return of 5p at the measured bound (r), and the best frequency over the calculation without excited atoms where that exists (f/f_0).

fill	p	n_e	f_{best}	T_e		x_{Kr^*}	p_{net}/P	k	$5p$	r	f/f_0
Ne	0.1	10^{14}	0.64	4.69	a	0.25	0.01	1.000	1.001	1.000	0.997
Ne	0.1	3×10^{14}	1.03	3.66	a	0.74	0.05	1.001	1.007	1.000	0.989
Ne	0.1	10^{15}	1.74	2.84	a	0.88	0.13	1.001	1.010	1.000	0.979
Ne	0.1	3×10^{15}	2.90	2.27	a	0.63	0.17	1.000	1.003	1.000	0.984
Ne	0.1	10^{16}	5.71	1.83	a	0.36	0.18	1.000	1.000	1.000	0.997
Ne	0.2	10^{14}	0.86	4.03	a	0.20	0.01	1.000	1.002	1.000	0.997
Ne	0.2	3×10^{14}	1.39	3.22	a	0.62	0.07	1.002	1.011	1.000	0.984
Ne	0.2	10^{15}	2.35	2.53	a	0.70	0.19	1.002	1.017	1.000	0.970
Ne	0.2	3×10^{15}	3.85	2.04	a	0.47	0.23	1.000	1.009	1.000	0.975
Ne	0.2	10^{16}	7.21	1.68	–	0.28	0.25	1.000	1.001	1.000	0.989
Ne	0.5	10^{14}	1.28	3.40	a	0.15	0.02	1.001	1.003	1.000	0.996
Ne	0.5	3×10^{14}	2.07	2.76	a	0.49	0.13	1.005	1.022	1.000	0.974
Ne	0.5	10^{15}	3.45	2.11	a	0.60	0.36	1.006	1.043	0.999	0.943
Ne	0.5	3×10^{15}	5.66	1.78	–	0.35	0.41	1.001	1.023	1.000	0.957
Ne	0.5	10^{16}	10.27	1.54	b	0.24	0.52	1.000	1.009	1.000	0.974
Ne	1	10^{14}	1.74	3.06	a	0.12	0.02	1.001	1.004	1.000	0.995
Ne	1	3×10^{14}	2.79	2.44	a	0.44	0.21	1.010	1.038	1.000	0.958
Ne	1	10^{15}	4.69	1.93	–	0.40	0.47	1.008	1.054	1.000	0.933
Ne	1	3×10^{15}	7.70	1.65	b	0.27	0.65	1.003	1.039	0.999	0.945
Ne	1	10^{16}	13.79	1.47	b	0.24	1.15	1.001	1.022	1.000	0.958
Ne	2	10^{14}	2.36	2.80	a	0.10	0.04	1.002	1.007	1.000	0.993
Ne	2	3×10^{14}	3.86	2.29	–	0.24	0.22	1.009	1.036	1.000	0.958
Ne	2	10^{15}	6.58	1.90	–	0.19	0.46	1.008	1.046	1.000	0.945
Ne	2	3×10^{15}	10.66	1.58	b	0.19	1.00	1.006	1.049	0.999	0.938
Ne	2	10^{16}	18.78	1.40	b	0.26	2.89	1.003	1.045	0.999	0.935
Ar	0.1	10^{14}	1.00	2.65	a	0.64	0.01	1.001	1.003	1.000	0.995
Ar	0.1	3×10^{14}	1.67	2.30	a	1.93	0.08	1.005	1.024	0.997	0.966
Ar	0.1	10^{15}	2.87	2.01	a	1.96	0.19	1.003	1.031	0.996	0.944
Ar	0.1	3×10^{15}	4.58	1.70	a	1.43	0.24	1.000	1.017	0.998	0.952
Ar	0.1	10^{16}	8.16	1.42	–	0.97	0.29	1.000	1.006	0.999	0.969
Ar	0.2	10^{14}	1.34	2.34	a	0.62	0.02	1.002	1.008	0.999	0.991
Ar	0.2	3×10^{14}	2.19	2.03	a	1.80	0.13	1.011	1.050	0.995	0.942
Ar	0.2	10^{15}	3.76	1.79	a	1.71	0.30	1.008	1.065	0.989	0.910
Ar	0.2	3×10^{15}	6.02	1.55	–	1.18	0.36	1.001	1.036	0.995	0.925
Ar	0.2	10^{16}	10.46	1.33	b	0.95	0.51	1.000	1.019	0.998	0.944
Ar	0.5	10^{14}	1.95	2.04	a	0.71	0.05	1.007	1.023	0.999	0.976
Ar	0.5	3×10^{14}	3.03	1.65	–	1.94	0.29	1.031	1.138	0.996	0.864

fill	p	n_e	f_{best}	T_e		x_{Kr^*}	p_{net}/P	k	$5p$	r	f/f_0
Ar	0.5	10^{15}	5.32	1.52	–	1.34	0.53	1.023	1.140	0.988	0.847
Ar	0.5	3×10^{15}	8.55	1.36	*b*	1.13	0.86	1.010	1.116	0.980	0.860
Ar	0.5	10^{16}	14.53	1.18	*b*	1.46	1.81	1.004	1.097	0.974	0.865
Ar	1	10^{14}	2.56	1.88	–	0.71	0.09	1.013	1.043	1.000	0.956
Ar	1	3×10^{14}	4.35	1.70	–	0.72	0.23	1.024	1.093	0.998	0.905
Ar	1	10^{15}	7.48	1.52	*b*	0.72	0.62	1.027	1.140	0.993	0.857
Ar	1	3×10^{15}	11.03	1.21	*b*	1.19	1.79	1.035	1.226	0.974	0.786
Ar	1	10^{16}	16.70	0.90	*b*	3.57	6.48	1.036	1.349	0.951	0.699
Ar	2	10^{14}	3.51	1.91	–	0.20	0.05	1.007	1.020	1.000	0.979
Ar	2	3×10^{14}	6.31	1.79	–	0.23	0.16	1.015	1.052	0.999	0.945
Ar	2	10^{15}	10.98	1.61	*b*	0.33	0.65	1.022	1.105	0.997	0.893
Ar	2	3×10^{15}	15.75	1.23	*b*	0.88	3.08	1.043	1.245	0.986	0.780
Ar	2	10^{16}	22.55	0.87	*b*	3.37	15.05	1.047	1.460	0.971	0.646
Kr	0.1	10^{14}	1.05	2.44	*a*	1.26	0.02	1.002	1.006	0.998	–
Kr	0.1	3×10^{14}	1.77	2.13	*a*	3.44	0.12	1.008	1.039	0.989	–
Kr	0.1	10^{15}	3.10	1.87	*a*	3.06	0.24	1.004	1.044	0.987	–
Kr	0.1	3×10^{15}	4.95	1.58	*a*	2.31	0.29	1.001	1.025	0.992	–
Kr	0.1	10^{16}	8.82	1.34	*b*	2.00	0.42	1.000	1.013	0.997	–
Kr	0.2	10^{14}	1.39	2.13	*a*	1.25	0.03	1.004	1.014	0.998	–
Kr	0.2	3×10^{14}	2.28	1.81	*a*	3.42	0.21	1.022	1.089	0.975	–
Kr	0.2	10^{15}	3.98	1.62	–	2.88	0.41	1.012	1.103	0.972	–
Kr	0.2	3×10^{15}	6.55	1.46	–	1.89	0.48	1.002	1.056	0.981	–
Kr	0.2	10^{16}	11.36	1.26	*b*	2.10	0.87	1.000	1.039	0.988	–
Kr	0.5	10^{14}	–	–	–	–	–	–	–	–	–
Kr	0.5	3×10^{14}	3.38	1.63	–	1.78	0.26	1.028	1.112	0.993	–
Kr	0.5	10^{15}	6.01	1.52	–	1.45	0.57	1.022	1.131	0.978	–
Kr	0.5	3×10^{15}	9.38	1.30	*b*	1.81	1.25	1.016	1.151	0.940	–
Kr	0.5	10^{16}	15.56	1.10	*b*	3.43	3.48	1.010	1.165	0.868	–
Kr	1	10^{14}	2.75	1.79	–	0.59	0.07	1.010	1.031	0.999	–
Kr	1	3×10^{14}	4.89	1.70	–	0.62	0.20	1.019	1.069	0.997	–
Kr	1	10^{15}	8.64	1.56	*b*	0.74	0.66	1.024	1.118	0.989	–
Kr	1	3×10^{15}	12.72	1.24	*b*	1.59	2.47	1.035	1.222	0.951	–
Kr	1	10^{16}	19.15	0.94	*b*	5.13	10.28	1.036	1.364	0.907	–
Kr	2	10^{14}	3.76	1.83	–	0.16	0.04	1.005	1.014	1.000	–
Kr	2	3×10^{14}	7.12	1.81	*b*	0.19	0.14	1.011	1.036	0.999	–
Kr	2	10^{15}	12.81	1.68	*b*	0.33	0.69	1.018	1.080	0.996	–
Kr	2	3×10^{15}	18.92	1.33	*b*	1.01	4.04	1.037	1.203	0.975	–
Kr	2	10^{16}	26.88	0.95	*b*	4.30	23.46	1.049	1.436	0.936	–

References

[1] G. J. M. Hagelaar and L. C. Pitchford, Solving the Boltzmann equation to obtain electron transport coefficients and rate coefficients for fluid models, Plasma Sources Sci. Technol. **14**, 722 (2005), doi:10.1088/0963-0252/14/4/011.

[2] E. Carbone et al., Data needs for modeling low-temperature non-equilibrium plasmas: the LXCat project, history, perspectives and a tutorial, Atoms **9**, 16 (2021), doi:10.3390/atoms9010016.

[3] D. Turnbull, J. Katz, M. Sherlock, L. Divol, N. R. Shaffer, D. J. Strozzi, A. Colaïtis, D. H. Edgell, R. K. Follett, K. R. McMillen, P. Michel, A. L. Milder and

D. H. Froula, Inverse bremsstrahlung absorption, Phys. Rev. Lett. **130**, 145103 (2023), doi:10.1103/PhysRevLett.130.145103.

[4] D. Turnbull, J. Katz, M. Sherlock, A. L. Milder, M. S. Cho, L. Divol, N. R. Shaffer, D. J. Strozzi, P. Michel and D. H. Froula, Reconciling calculations and measurements of inverse bremsstrahlung absorption, Phys. Plasmas **31**, 063304 (2024), doi:10.1063/5.0203446.

[5] O. Klein and S. Rosseland, Über Zusammenstöße zwischen Atomen und freien Elektronen, Z. Phys. **4**, 46 (1921), doi:10.1007/BF01328041.

[6] A. Kramida, Yu. Ralchenko, J. Reader and NIST ASD Team, NIST Atomic Spectra Database, NIST Standard Reference Database 78, doi:10.18434/T4W30F, Kr I, Ar I and Ne I levels read October 8, 2026.

[7] F. Kannari, M. Obara and T. Fujioka, An advanced kinetic model of electron-beam-excited KrF lasers including the vibrational relaxation in KrF*(B) and collisional mixing of KrF*(B,C), J. Appl. Phys. **57**, 4309 (1985), doi:10.1063/1.334590.

[8] T. H. Johnson and A. M. Hunter, Physics of the krypton fluoride laser, J. Appl. Phys. **51**, 2406 (1980), doi:10.1063/1.328010.

[9] L. Vriens, A semi-empirical formula for calculation of absolute cross sections for ionization and excitation of atoms by electrons, Physica **31**, 385 (1965), doi:10.1016/0031-8914(65)90043-1.

[10] J. H. Jacob and J. A. Mangano, Modeling the KrF laser discharge, Appl. Phys. Lett. **28**, 724 (1976), doi:10.1063/1.88629.

[11] R. O. Jung, T. E. Stone, J. B. Boffard, L. W. Anderson and C. C. Lin, Electron-impact excitation out of the metastable levels of krypton, Phys. Rev. Lett. **94**, 163202 (2005), doi:10.1103/PhysRevLett.94.163202.

[12] R. O. Jung, T. E. Stone, J. B. Boffard, L. W. Anderson and C. C. Lin, Electronic structure and angular momentum coupling as reflected in electron excitation out of rare-gas metastable levels, Phys. Rev. A **73**, 022722 (2006), doi:10.1103/PhysRevA.73.022722.

[13] J. E. Velazco, J. H. Kolts and D. W. Setser, Quenching rate constants for metastable argon, krypton, and xenon atoms by fluorine containing molecules and branching ratios for XeF* and KrF* formation, J. Chem. Phys. **65**, 3468 (1976), doi:10.1063/1.433573.

[14] K. Ueda, S. Kanada, A. Sasaki and H. Takuma, A pulse radiolysis study of KrF laser kinetics (II), Rev. Laser Eng. **12**, 357 (1984), doi:10.2184/lsj.12.357.

[15] Y. C. Yu and D. W. Setser, Krypton chloride and krypton fluoride thermochemistry and formation and relaxation kinetics in argon, nitrogen, and tetrafluoromethane buffer gas, J. Phys. Chem. **94**, 2934 (1990), doi:10.1021/j100370a037.

[16] W. Wieme and J. Lenaerts, Excimer formation in argon, krypton, and xenon discharge afterglows between 200 and 400 K, J. Chem. Phys. **74**, 483 (1981), doi:10.1063/1.440855.

[17] E. Morikawa, R. Reininger, P. Gürtler, V. Saile and P. Laporte, Argon, krypton, and xenon excimer luminescence: from the dilute gas to the condensed phase, J. Chem. Phys. **91**, 1469 (1989), doi:10.1063/1.457108.

[18] G. P. Quigley and W. M. Hughes, Lifetime and quenching rate constants for Kr_2F^* and Kr_2^*, Appl. Phys. Lett. **32**, 649 (1978), doi:10.1063/1.89882.

[19] C. Y. Ng, D. J. Trevor, B. H. Mahan and Y. T. Lee, Photoionization studies of the Kr_2 and Ar_2 van der Waals molecules, J. Chem. Phys. **66**, 446 (1977), doi:10.1063/1.433989.

[20] J. W. Keto, R. E. Gleason and G. K. Walters, Production mechanisms and radiative lifetimes of argon and xenon molecules emitting in the ultraviolet, Phys. Rev. Lett. **33**, 1365 (1974), doi:10.1103/PhysRevLett.33.1365.

[21] C. H. Chen and M. G. Payne, Kinetics of proton-excited Ar–F_2 and Kr–F_2 mixtures, IEEE J. Quantum Electron. **15**, 149 (1979), doi:10.1109/JQE.1979.1069984.

[22] D. W. Setser, Experimental investigation of the chemistry of excited states of rare gases, annual technical progress report for October 15, 1976 to October 14, 1977, report COO-2807-11 (1978), doi:10.2172/6681231.

[23] Y.-W. Lee, E. Matsui, F. Kannari and M. Obara, Theoretical high-efficiency extraction study of a short-pulse electron-beam-pumped ArF laser amplifier with atmospheric pressure Ar-rich mixtures, IEEE J. Quantum Electron. **25**, 2053 (1989), doi:10.1109/3.35232.

[24] S. Rauf and M. J. Kushner, Dynamics of a coplanar-electrode plasma display panel cell. I. Basic operation, J. Appl. Phys. **85**, 3460 (1999), doi:10.1063/1.369703.

[25] J. Chen, L. Zhang, Z. Zhao and Y. Fu, Fluid modeling of discharge-pumped ArF excimer lasers: effects of F_2 concentration on pulsed discharge dynamics, Phys. Plasmas **32**, 113502 (2025), doi:10.1063/5.0287524.

[26] X. Ma, L. Bai, Y. Zhu, X. Jiang and Y. Wu, Numerical investigation of discharge evolution and breakdown characteristics of ArF excimer lasers, Plasma Sources Sci. Technol. **33**, 075012 (2024), doi:10.1088/1361-6595/ad5df6.

[27] A. M. Boichenko and S. I. Yakovlenko, The possibility of lasing in Ne^+Ar ionic molecules pumped by a hard ioniser, Quantum Electron. **30**, 681 (2000), doi:10.1070/QE2000v030n08ABEH001791.

[28] K. Gazeli, X. Aubert, S. Prasanna, C. Y. Duluard, G. Lombardi and K. Hassouni, Picosecond two-photon absorption laser-induced fluorescence (ps-TALIF) in krypton: the role of photoionization on the density depletion of the fluorescing state Kr $5p'[3/2]_2$, Phys. Plasmas **28**, 043301 (2021), doi:10.1063/5.0041471.

[29] K. Dzierżęga, U. Volz, G. Nave and U. Griesmann, Accurate transition rates for the 5p–5s transitions in Kr I, Phys. Rev. A **62**, 022505 (2000), doi:10.1103/PhysRevA.62.022505.

[30] C. A. Whitehead, H. Pournasr, M. R. Bruce, H. Cai, J. Kohel, W. B. Layne and J. W. Keto, Deactivation of two-photon excited Xe($5p^5 6p$, $6p'$, 7p) and Kr($4p^5 5p$) in xenon and krypton, J. Chem. Phys. **102**, 1965 (1995), doi:10.1063/1.468763.